

APEXPLANET SOFTWARE PVT. LTD.

45-Day Data Analytics Internship

FINAL PROJECT REPORT

Customer Churn Analysis using IBM Telco Customer Churn Dataset



Submitted By

Name: Guttula Sai Priya

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Project Title

Customer Churn Analysis, Business Intelligence Dashboard, Statistical Modeling and Machine Learning using the IBM Telco Customer Churn Dataset

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SQLite, SQLAlchemy, Power BI, Scikit-learn, Statsmodels, Jupyter Notebook

Declaration

I hereby declare that this project report entitled “**Customer Churn Analysis using the IBM Telco Customer Churn Dataset**” is the result of work carried out by me during the **45-Day Data Analytics Internship at ApexPlanet Software Pvt. Ltd.**

This report has been prepared for academic and internship learning purposes. All datasets, software tools, libraries, and references used in this project have been appropriately acknowledged. The analysis, visualizations, SQL queries, dashboard development, statistical modeling, and machine learning implementation presented in this report were completed as part of the internship activities.

I further declare that this report has not been submitted previously for any other internship, degree, diploma, or certification.

Student Name: Guttula Sai Priya

Place: Hyderabad

Date: 10/7/2026

Signature: G.SAI PRIYA

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Executive Summary

Customer retention is one of the most important challenges faced by organizations in highly competitive industries. Understanding why customers discontinue a service enables businesses to improve customer satisfaction, reduce revenue loss, and design effective retention strategies. This project focuses on analyzing customer churn using the **IBM Telco Customer Churn Dataset** through a complete end-to-end data analytics workflow.

The project was carried out as part of the **45-Day Data Analytics Internship at ApexPlanet Software Pvt. Ltd.** and demonstrates the practical application of data analytics, business intelligence, statistical analysis, and machine learning techniques. The primary objective was to transform raw customer data into meaningful business insights that support data-driven decision-making.

The project began with setting up a structured analytics environment and performing comprehensive data cleaning and preprocessing. Missing values were identified and handled appropriately, duplicate records were removed, data types were standardized, categorical variables were formatted correctly, and outlier detection techniques were applied to improve data quality. These preprocessing steps ensured that the dataset was reliable for subsequent analysis.

Exploratory Data Analysis (EDA) was performed using Python libraries including Pandas, Matplotlib, Seaborn, and Plotly. Multiple visualizations were created to examine customer demographics, contract types, payment methods, internet services, monthly charges, tenure, and churn behavior. Correlation analysis and interactive visualizations helped identify important relationships among different variables and highlighted patterns influencing customer churn.

The processed data was then integrated into a SQLite database to perform SQL-based business analysis. SQL queries involving filtering, aggregation, grouping, subqueries, Common Table Expressions (CTEs), and window functions were developed to extract valuable business information. Python and SQLAlchemy were integrated to automate data retrieval and simplify analytical workflows.

To support business decision-making, an interactive Power BI dashboard was developed. The dashboard includes KPI cards, churn distribution charts, contract analysis, payment method comparisons, internet service analysis, scatter plots, matrices, and interactive slicers. These visualizations allow users to explore customer behavior dynamically and monitor important business metrics.

Advanced analytics techniques were also applied during the project. A time-series analysis was demonstrated by constructing a monthly trend using customer tenure and average monthly charges. Seasonal decomposition, stationarity testing using the Augmented Dickey-Fuller (ADF) test, moving averages, exponential smoothing, and ARIMA forecasting were implemented to illustrate forecasting concepts. Customer segmentation was performed using the K-Means clustering algorithm after feature standardization, with the optimal number of clusters identified using the Elbow Method and Silhouette Score. Principal Component Analysis (PCA) was used to visualize the customer segments in two dimensions.

Predictive modeling was performed using multiple machine learning algorithms. Linear Regression was used to predict the numerical variable **MonthlyCharges**, while Logistic Regression and Decision Tree Classification were used to predict customer churn. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, MAE, RMSE, and R^2 metrics. Additional evaluation techniques such as confusion matrices, ROC curves, and feature importance analysis were used to assess model performance and interpret the results.

The analysis revealed several important business insights. Customers with month-to-month contracts showed significantly higher churn rates than those with long-term contracts. Customers paying higher

monthly charges were generally more likely to discontinue their services. Longer-tenure customers demonstrated stronger loyalty and lower churn rates. Internet service type and payment method also influenced customer retention patterns. These findings provide practical recommendations for improving customer retention through targeted engagement strategies, contract optimization, personalized offers, and proactive customer support.

Overall, this project demonstrates a complete data analytics lifecycle, beginning with data collection and preprocessing and extending through exploratory analysis, SQL querying, dashboard development, statistical modeling, customer segmentation, predictive analytics, and business recommendations. It highlights the effective use of modern analytics tools and techniques to solve real-world business problems and supports informed decision-making through data-driven insights.

Problem Statement & Project Objectives

1. Problem Statement

In today's highly competitive telecommunications industry, retaining existing customers is as important as acquiring new ones. Customer churn, the phenomenon where customers discontinue using a company's services, directly impacts revenue, customer lifetime value, and overall business growth. Organizations invest significant resources in customer acquisition, making it essential to understand the factors that influence customer retention and satisfaction.

The IBM Telco Customer Churn dataset contains customer demographic information, service subscriptions, account details, billing information, and churn status. Although this data provides valuable information, extracting meaningful insights from raw data requires systematic data analysis and visualization techniques. Businesses need efficient analytical solutions that can identify churn patterns, understand customer behavior, and support strategic decision-making.

This project addresses the challenge of transforming raw customer data into actionable business insights through data cleaning, exploratory analysis, SQL-based querying, dashboard development, statistical analysis, customer segmentation, and predictive modeling. The findings can help businesses identify customers who are at a higher risk of churning and develop targeted retention strategies.

2. Project Objectives

The primary objective of this project is to perform a complete end-to-end data analytics workflow on the IBM Telco Customer Churn dataset and generate meaningful insights that support business decision-making.

The specific objectives of the project are:

- To establish a structured data analytics environment using Python, Jupyter Notebook, SQLite, and Power BI.
- To collect and understand the IBM Telco Customer Churn dataset, including its attributes, data types, and business context.
- To clean and preprocess the dataset by handling missing values, removing duplicates, correcting data types, standardizing column names, and identifying outliers.
- To perform Exploratory Data Analysis (EDA) using statistical summaries and graphical visualizations to identify trends, distributions, and relationships among variables.
- To store the cleaned dataset in a relational database and perform SQL-based data extraction using filtering, grouping, aggregation, subqueries, Common Table Expressions (CTEs), and window functions.
- To integrate Python with SQLite using SQLAlchemy for automated data retrieval and analysis.
- To build interactive dashboards in Power BI for monitoring key performance indicators (KPIs) and customer behavior.
- To perform statistical analysis and demonstrate time-series forecasting techniques using seasonal decomposition, moving averages, exponential smoothing, and ARIMA models.
- To segment customers into meaningful groups using the K-Means clustering algorithm and visualize the clusters using Principal Component Analysis (PCA).

- To develop predictive models using Linear Regression, Logistic Regression, and Decision Tree algorithms and evaluate their performance using appropriate statistical metrics.
- To provide actionable business recommendations based on analytical findings to improve customer retention and support data-driven decision-making.
- To document the complete analytical workflow professionally for GitHub, internship submission, and future portfolio development.

3. Scope of the Project

This project focuses on analyzing customer churn using the IBM Telco Customer Churn dataset through modern data analytics and machine learning techniques. The scope includes data preprocessing, exploratory analysis, SQL querying, interactive dashboard development, statistical modeling, customer segmentation, predictive analytics, and business reporting.

The project does not involve collecting primary data or deploying a production-level machine learning application. Instead, it demonstrates the practical application of industry-standard analytical tools and techniques for solving a real-world business problem using an existing dataset.

4. Expected Outcomes

The expected outcomes of this project include:

- A cleaned and well-structured customer dataset suitable for analysis.
- Interactive visualizations and dashboards for business decision-making.
- SQL queries that extract meaningful business insights.
- Statistical models that demonstrate forecasting techniques.
- Customer segments identified through clustering analysis.
- Predictive models capable of estimating customer churn and numerical outcomes.
- Business recommendations to improve customer retention strategies.
- A professional project repository containing notebooks, reports, dashboards, automation scripts, and documentation.

The successful completion of this project demonstrates the practical application of data analytics, business intelligence, statistical modeling, and machine learning concepts in solving real-world business challenges.

Dataset Description & Methodology

1. Dataset Description

1.1 Dataset Overview

This project utilizes the **IBM Telco Customer Churn Dataset**, a widely recognized public dataset frequently used for data analytics, business intelligence, and machine learning applications. The dataset represents customer information collected from a fictional telecommunications company and is designed to analyze the factors that influence customer churn.

Customer churn refers to the situation in which a customer stops using the company's services. Understanding the reasons behind customer churn enables organizations to improve customer satisfaction, reduce revenue loss, and develop effective customer retention strategies.

The dataset contains customer demographic details, subscription information, billing details, service usage, payment methods, contract information, and the customer's churn status. These attributes provide a comprehensive view of customer behavior and enable meaningful business analysis.

1.2 Data Source

Dataset Name: IBM Telco Customer Churn Dataset

Source: Kaggle – IBM Sample Data

Dataset Link: <https://www.kaggle.com/datasets/blaschar/telco-customer-churn>

Dataset Type: Structured CSV Dataset

File Format: CSV (Comma-Separated Values)

Domain: Telecommunications

Primary Business Problem: Customer Churn Prediction and Customer Retention Analysis

1.3 Dataset Characteristics

The dataset consists of customer records containing demographic information, subscribed services, billing details, and customer churn information.

Key Characteristics:

- Structured tabular dataset
- Combination of numerical and categorical variables
- Suitable for classification and regression tasks
- Suitable for SQL querying and dashboard development
- Appropriate for clustering and customer segmentation
- Supports business intelligence reporting

1.4 Dataset Features

The major attributes used in this project are described below.

Column	Description
CustomerID	Unique identifier assigned to each customer
Gender	Customer gender (Male/Female)
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Indicates whether the customer has a partner
Dependents	Indicates whether the customer has dependents
Tenure	Number of months the customer has stayed with the company
PhoneService	Availability of phone service
MultipleLines	Whether multiple phone lines are subscribed
InternetService	Type of internet service (DSL, Fiber Optic, No Internet)
OnlineSecurity	Availability of online security service
OnlineBackup	Availability of online backup service
DeviceProtection	Device protection subscription
TechSupport	Technical support subscription
StreamingTV	Streaming TV subscription
StreamingMovies	Streaming Movies subscription
Contract	Contract type (Month-to-Month, One Year, Two Year)
PaperlessBilling	Paperless billing status
PaymentMethod	Customer payment method
MonthlyCharges	Monthly bill amount
TotalCharges	Total amount paid by the customer
Churn	Target variable indicating whether the customer left the company

1.5 Data Collection Method

The IBM Telco Customer Churn dataset is a publicly available sample dataset created for educational and analytical purposes. It represents customer records collected from a telecommunications business environment and includes information relevant to customer behavior, billing patterns, and service usage.

Although the dataset is not collected directly from a live production environment, it closely resembles real-world business data and is widely adopted for learning data analytics, predictive modeling, and customer segmentation techniques.

1.6 Dataset Limitations

Although the dataset is comprehensive, several limitations should be considered during analysis.

- The dataset represents a sample rather than real-time business operations.
- Customer information is limited to the available variables.
- External business factors such as market competition, customer satisfaction surveys, and promotional campaigns are not included.
- The dataset does not contain actual transaction timestamps, limiting true time-series forecasting.
- Geographic information is limited, restricting regional analysis.
- The dataset represents a snapshot of customer information rather than continuous historical records.

These limitations should be considered when interpreting analytical results and business recommendations.

2. Project Methodology

A structured data analytics methodology was followed throughout the project to ensure accurate analysis and meaningful business insights.

The complete workflow consisted of multiple sequential stages, beginning with data acquisition and ending with predictive modeling and reporting.

Phase 1 – Data Collection

The IBM Telco Customer Churn dataset was downloaded from Kaggle and stored within the project directory.

The dataset was imported into Python using the Pandas library for further processing.

Phase 2 – Environment Setup

The analytics environment was configured using Python and Jupyter Notebook.

The following libraries were installed and used throughout the project:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- SQLAlchemy
- SQLite
- Scikit-learn
- Statsmodels
- Joblib

A structured project folder hierarchy was created to organize datasets, notebooks, SQL scripts, reports, dashboards, visualizations, models, and automation scripts.

Phase 3 – Data Cleaning and Preprocessing

Data preprocessing involved several important steps to improve data quality.

These included:

- Identifying missing values
- Converting data types
- Handling invalid values
- Removing duplicate records
- Standardizing column names
- Detecting outliers using the IQR method
- Encoding categorical variables where required
- Saving the cleaned dataset for subsequent analysis

The cleaned dataset served as the foundation for all later analytical tasks.

Phase 4 – Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behavior and identify meaningful patterns.

Several statistical summaries and visualizations were created, including:

- Histograms
- Boxplots
- Correlation Heatmaps
- Pairplots
- Scatter Plots
- Count Plots
- Pie Charts
- Distribution Plots

EDA helped identify relationships among customer tenure, monthly charges, contract types, internet services, payment methods, and churn.

Phase 5 – SQL Data Analysis

The cleaned dataset was imported into a SQLite database.

SQL queries were developed to perform business analysis using:

- SELECT
- WHERE
- ORDER BY
- LIMIT
- GROUP BY
- HAVING
- Aggregate Functions

- Subqueries
- Common Table Expressions (CTEs)
- Window Functions

Python was integrated with SQLite using SQLAlchemy to automate query execution and data retrieval.

Phase 6 – Dashboard Development

Interactive dashboards were developed using Power BI.

The dashboard includes:

- KPI Cards
- Customer Distribution
- Contract Analysis
- Churn Analysis
- Internet Service Analysis
- Payment Method Analysis
- Scatter Charts
- Tables and Matrices
- Interactive Slicers

The dashboard enables users to explore customer information interactively and supports business decision-making.

Phase 7 – Statistical Analysis

Advanced analytical techniques were implemented using Statsmodels.

The following analyses were performed:

- Time-series creation
- Seasonal decomposition
- Stationarity testing using the Augmented Dickey-Fuller (ADF) Test
- Moving Average
- Exponential Smoothing
- ARIMA Forecasting

Forecast performance was evaluated using MAE and RMSE metrics.

Phase 8 – Customer Segmentation

Customer segmentation was performed using the K-Means clustering algorithm.

The workflow included:

- Feature selection
- Data standardization
- Elbow Method
- Silhouette Score evaluation

- Cluster creation
- Principal Component Analysis (PCA)
- Cluster visualization

The identified customer segments provide insights into different customer groups based on tenure, billing behavior, and spending patterns.

Phase 9 – Predictive Modeling

Machine learning models were developed to predict customer behavior.

The models implemented include:

- Linear Regression (Monthly Charges Prediction)
- Logistic Regression (Customer Churn Prediction)
- Decision Tree Classification

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- MAE
- RMSE
- R² Score

Feature importance analysis, confusion matrices, and ROC curves were also generated to improve model interpretability.

Phase 10 – Reporting and Deployment

The final phase involved preparing a complete project repository and documentation.

The project deliverables include:

- Jupyter Notebooks
- SQL Scripts
- Power BI Dashboard
- Interactive Visualizations
- Machine Learning Models
- Automation Pipeline
- Executive Report
- GitHub Repository
- Presentation Slides
- Demonstration Video

This structured methodology ensured that every stage of the analytics lifecycle was properly documented and implemented, resulting in a comprehensive end-to-end data analytics project suitable for business analysis, academic learning, and professional portfolio development.

Data Cleaning & Preprocessing

1. Introduction

Data cleaning and preprocessing are fundamental stages of every data analytics project. Real-world datasets often contain missing values, inconsistent data types, duplicate records, invalid entries, and other quality issues that can negatively impact analysis and machine learning models.

The objective of this phase was to prepare the IBM Telco Customer Churn dataset for accurate analysis by improving data quality, ensuring consistency, and transforming the data into a format suitable for statistical analysis, visualization, SQL querying, dashboard development, and predictive modeling.

The preprocessing operations were performed using Python libraries such as Pandas and NumPy within the Jupyter Notebook environment.

2. Initial Data Inspection

Before applying any transformations, the dataset was carefully inspected to understand its structure and identify potential quality issues.

The following operations were performed:

- Loaded the CSV dataset into a Pandas DataFrame.
- Displayed the first few records using `head()`.
- Examined dataset dimensions using `shape`.
- Reviewed column names.
- Verified data types using `info()`.
- Generated descriptive statistics using `describe()`.
- Checked for missing values.
- Identified duplicate records.

This initial inspection provided a clear understanding of the dataset and guided the subsequent preprocessing steps.

`df.head()`

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	—	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1	5575-GNVBE	Male	0	No	No	34	Yes	No	DSL	Yes	—	Yes	No	No	One year	No	Mailed check	56.95	1889.5	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	—	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	—	Yes	Yes	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure                7043 non-null   int64
6   PhoneService          7043 non-null   object
7   Multiplelines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

3. Missing Value Handling

Handling missing values is essential because incomplete records can reduce the reliability of analytical results.

During inspection, the TotalCharges column contained blank values that were interpreted as missing after converting the column to a numeric data type.

The following preprocessing steps were performed:

- Converted the TotalCharges column from text to numeric format.
- Invalid values were automatically converted to NaN.
- Missing values were replaced using the median value of the column.

The median was selected because it is less sensitive to extreme values and provides a robust estimate for numerical data.

Advantages

- Prevented loss of customer records.
- Improved dataset completeness.
- Preserved statistical characteristics of the data.

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines    0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
dtype: int64
```

4. Duplicate Record Removal

Duplicate records may introduce bias into statistical analysis and machine learning models by counting the same observation multiple times.

The dataset was examined using the `duplicated()` function.

Duplicate entries, if present, were removed using the `drop_duplicates()` method.

Benefits

- Eliminated redundant observations.
- Improved dataset accuracy.
- Prevented duplicate customer records from influencing analytical results.

```
df.duplicated().sum()

np.int64(0)
```

5. Data Type Conversion

Correct data types are necessary for efficient computation and accurate analysis.

Several transformations were performed during preprocessing.

These included:

- Converting TotalCharges from object type to numeric.
- Converting categorical variables to the category data type.
- Ensuring numerical columns remained in numeric format.

Proper data typing improves memory efficiency and simplifies statistical analysis.

Benefits

- Improved computational performance.
- Reduced memory consumption.
- Simplified visualization and modeling.

6. Standardization of Column Names

To improve readability and maintain consistency throughout the project, all column names were standardized.

The preprocessing included:

- Converting all column names to lowercase.
- Removing inconsistencies in naming conventions.

Example:

Before:

- CustomerID
- MonthlyCharges
- TotalCharges

After:

- customerid
- monthlycharges
- totalcharges

Standardized naming simplifies SQL queries, Python scripts, and dashboard development.

7. Outlier Detection

Outliers are observations that significantly differ from the majority of the data.

The Monthly Charges variable was analyzed using the Interquartile Range (IQR) method.

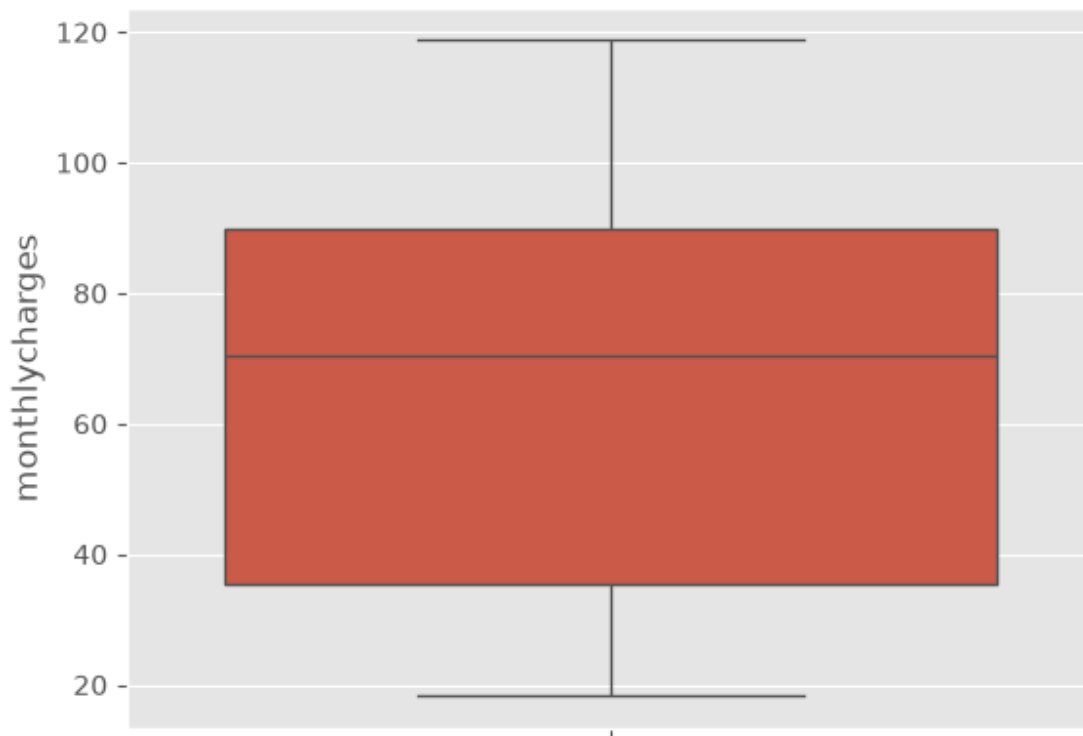
The following calculations were performed:

- First Quartile (Q1)

- Third Quartile (Q3)
- Interquartile Range (IQR)
- Lower Bound = $Q1 - 1.5 \times IQR$
- Upper Bound = $Q3 + 1.5 \times IQR$

Potential outliers were identified but were not removed because they represented valid customer billing amounts rather than data entry errors.

Retaining genuine observations preserves the integrity of customer behavior analysis.



8. Feature Transformation

Several feature transformations were performed to prepare the dataset for machine learning and statistical analysis.

The transformations included:

- Converting categorical variables into category data types.
- Encoding categorical variables during machine learning using Label Encoding.
- Preparing numerical features for clustering using StandardScaler.
- Saving the cleaned dataset for future tasks.

These transformations ensured compatibility with regression, classification, clustering, and forecasting algorithms.

9. Data Validation

After completing preprocessing, the dataset was validated to ensure consistency.

Validation activities included:

- Verifying that missing values had been handled.
- Confirming duplicate records had been removed.
- Checking corrected data types.
- Reviewing descriptive statistics.
- Confirming successful export of the cleaned dataset.

The final dataset was saved as:

data/processed/telco_cleaned.csv

This cleaned dataset was used throughout Tasks 2, 3, and 4 of the internship.

10. Data Cleaning Summary

The following preprocessing operations were completed successfully:

Data Cleaning Activity	Status
Dataset Loaded	Completed
Initial Data Inspection	Completed
Missing Values Identified	Completed
Missing Values Treated	Completed
Duplicate Records Removed	Completed
Data Types Corrected	Completed
Column Names Standardized	Completed
Outlier Detection	Completed
Feature Transformation	Completed
Cleaned Dataset Saved	Completed

11. Cleaning Log

The complete sequence of transformations performed during preprocessing is summarized below:

- Imported the IBM Telco Customer Churn dataset into Pandas.
- Examined dataset structure and column information.

- Identified missing values in the TotalCharges column.
- Converted TotalCharges to numeric format.
- Replaced missing values with the median value.
- Checked for duplicate customer records.
- Removed duplicate entries where applicable.
- Standardized all column names to lowercase.
- Converted categorical columns to the category data type.
- Performed outlier detection using the IQR method.
- Validated the processed dataset.
- Exported the cleaned dataset for SQL analysis, visualization, dashboarding, statistical analysis, and machine learning.

12. Outcome of Data Preprocessing

The preprocessing phase successfully transformed the raw customer dataset into a clean, consistent, and analysis-ready dataset.

The cleaned dataset served as the foundation for all subsequent stages of the project, including:

- Exploratory Data Analysis (EDA)
- SQL Data Extraction
- Power BI Dashboard Development
- Statistical Analysis
- Time-Series Forecasting
- Customer Segmentation
- Machine Learning Models

By ensuring high-quality input data, the preprocessing stage significantly improved the reliability, accuracy, and interpretability of the analytical results presented throughout this project.

Exploratory Data Analysis (EDA)

1. Introduction

Exploratory Data Analysis (EDA) is a critical stage of the data analytics process that focuses on understanding the structure, characteristics, and relationships within a dataset before applying statistical or machine learning techniques. The primary objective of EDA is to discover meaningful patterns, identify anomalies, validate assumptions, and generate business insights through statistical summaries and visualizations.

For this project, EDA was performed using Python libraries including Pandas, Matplotlib, Seaborn, and Plotly. Various statistical measures and graphical techniques were used to analyze customer demographics, billing information, subscribed services, contract types, and customer churn behavior.

The insights obtained during this phase guided subsequent SQL analysis, dashboard development, customer segmentation, and predictive modeling.

2. Statistical Summary

The first step in EDA involved generating descriptive statistics using the `describe()` function.

The statistical summary provided valuable information about numerical variables such as:

- Customer tenure
- Monthly charges
- Total charges

The summary included:

- Count
- Mean
- Standard Deviation
- Minimum Value
- Maximum Value
- 25th Percentile
- Median (50th Percentile)
- 75th Percentile

These statistics provided a comprehensive overview of customer spending patterns and service duration.

```
df.describe()
```

	seniorcitizen	tenure	monthlycharges	totalcharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2281.916928
std	0.368612	24.559481	30.090047	2265.270398
min	0.000000	0.000000	18.250000	18.800000
25%	0.000000	9.000000	35.500000	402.225000
50%	0.000000	29.000000	70.350000	1397.475000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

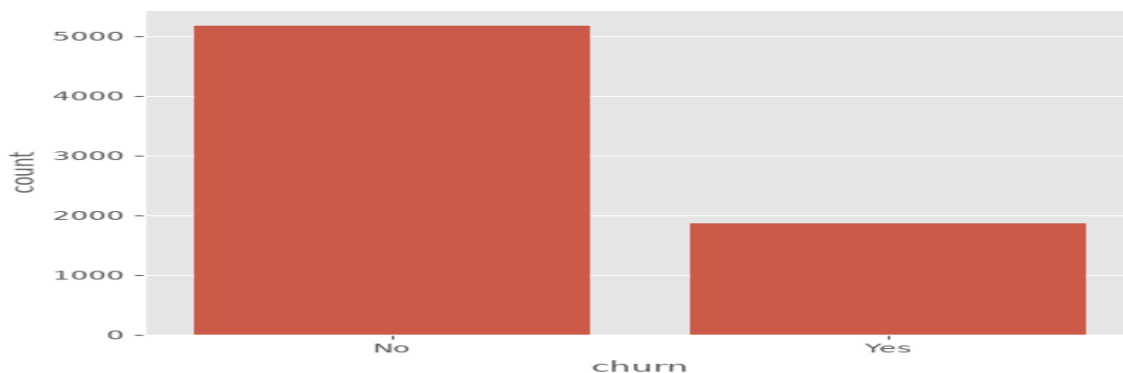
3. Customer Churn Distribution

A count plot was created to understand the distribution of customers who stayed with the company compared to those who discontinued the service.

The visualization revealed that the majority of customers remained with the company, while a smaller proportion had churned. Although the dataset is imbalanced toward retained customers, the number of churned customers is still substantial enough to warrant detailed analysis.

Observation

- Most customers continued using the company's services.
- A significant number of customers had discontinued their subscriptions.
- Customer churn remains an important business concern requiring proactive retention strategies.



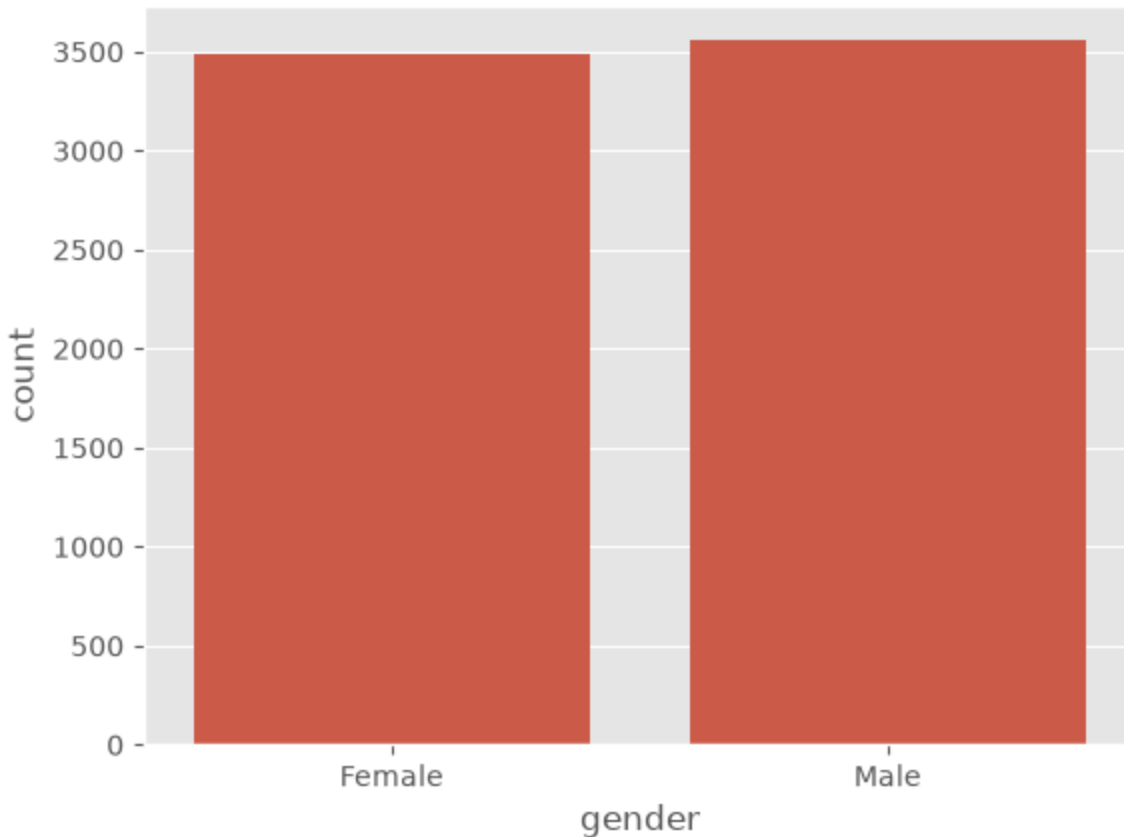
4. Gender Distribution

The gender distribution was analyzed to determine whether customer demographics were balanced.

The visualization indicated a nearly equal distribution of male and female customers, suggesting that the dataset does not exhibit significant gender bias.

Observation

- Male and female customers are represented almost equally.
- Gender alone does not appear to be a major factor influencing customer churn.
- Additional variables should be analyzed alongside gender to identify churn patterns.



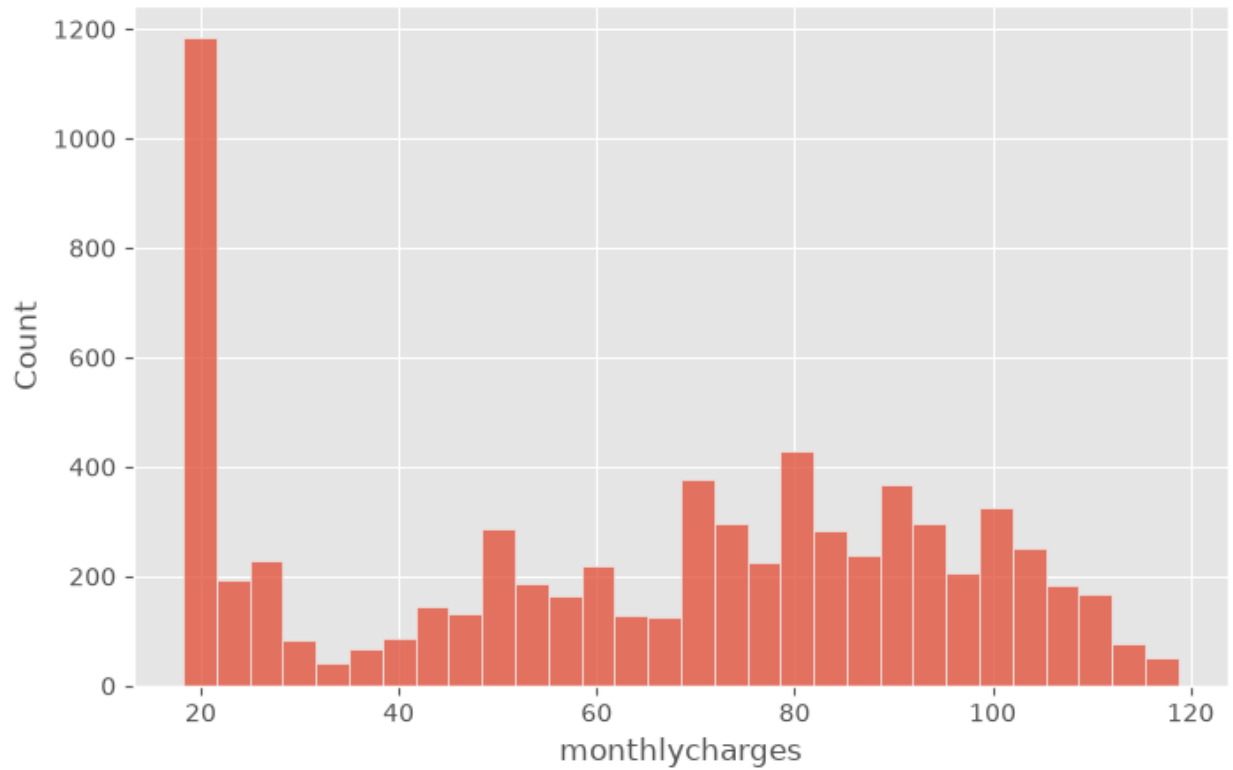
5. Monthly Charges Distribution

A histogram was used to examine the distribution of monthly charges across customers.

The analysis showed that customer monthly charges vary considerably depending on subscribed services and contract plans.

Observation

- Monthly charges are distributed across a broad range of values.
- Certain pricing levels contain a higher concentration of customers.
- Customers paying higher monthly charges require further investigation to understand their churn behavior.



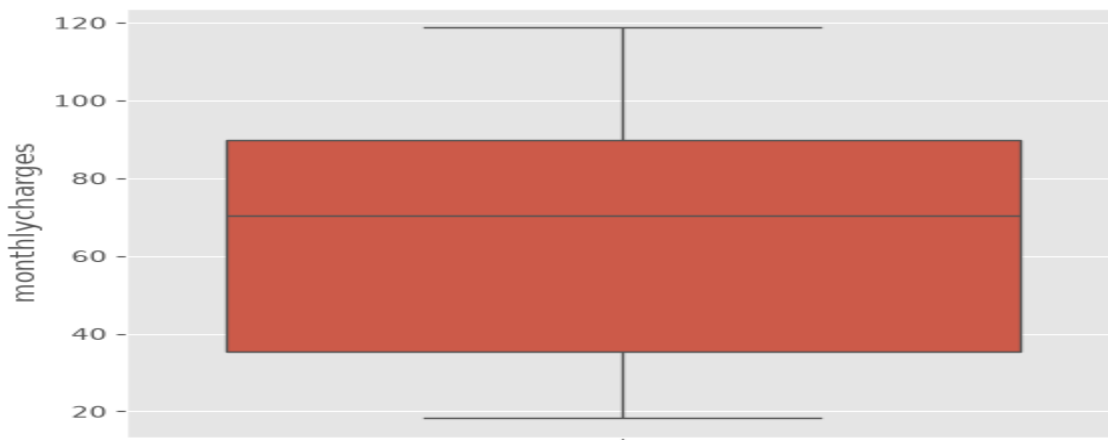
6. Outlier Analysis

A box plot was generated for the Monthly Charges variable to detect potential outliers.

The analysis indicated the presence of observations outside the typical range. However, these values represent genuine customer billing amounts rather than data entry errors.

Observation

- A few customers pay significantly higher monthly charges.
- These observations are valid business records.
- Outliers were retained because they provide meaningful analytical information.



7. Correlation Analysis

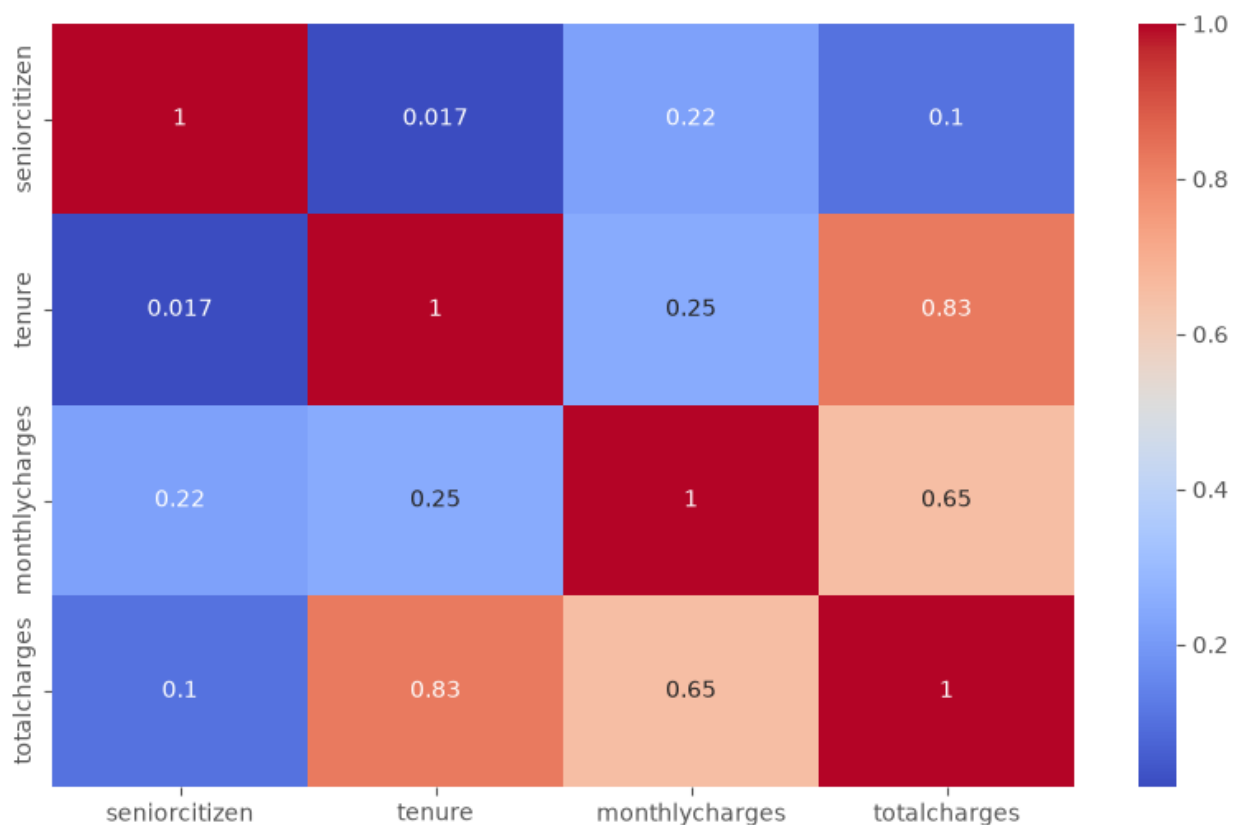
A correlation heatmap was created to examine relationships among numerical variables.

The heatmap illustrates the strength and direction of linear relationships between variables using correlation coefficients.

Observation

- Total Charges exhibit a strong positive relationship with customer tenure.
- Monthly Charges show a moderate relationship with Total Charges.
- No strong negative correlations were observed among numerical variables.

Understanding these relationships supports feature selection for predictive modeling.



8. Scatter Plot Analysis

A scatter plot was generated using customer tenure and monthly charges, with customer churn represented by color.

The visualization helps identify whether spending patterns and service duration influence customer churn.

Observation

- Customers with shorter tenure appear more likely to churn.
- Customers with higher monthly charges also show relatively higher churn.
- Long-term customers generally demonstrate stronger retention.

This visualization highlights important relationships that are later confirmed through machine learning models.



9. Contract Type Analysis

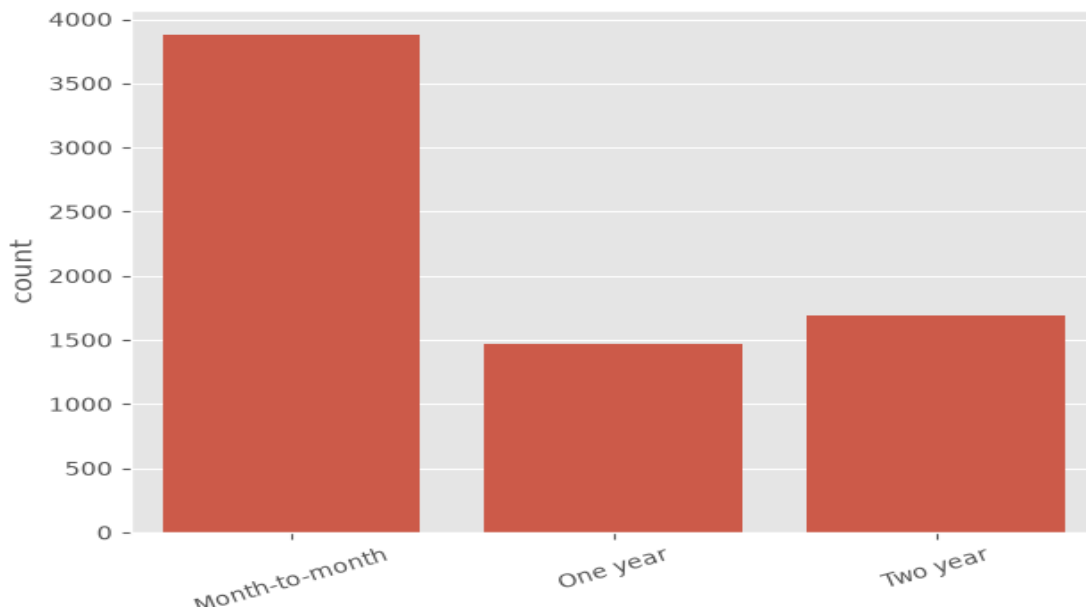
Customer contract types were analyzed to understand their relationship with churn.

The distribution of Month-to-Month, One-Year, and Two-Year contracts was examined.

Observation

- Month-to-Month contracts represent the largest customer group.
- Customers with Month-to-Month contracts exhibit noticeably higher churn.
- Long-term contracts are associated with improved customer retention.

Contract duration appears to be one of the strongest indicators of customer loyalty.



10. Internet Service Analysis

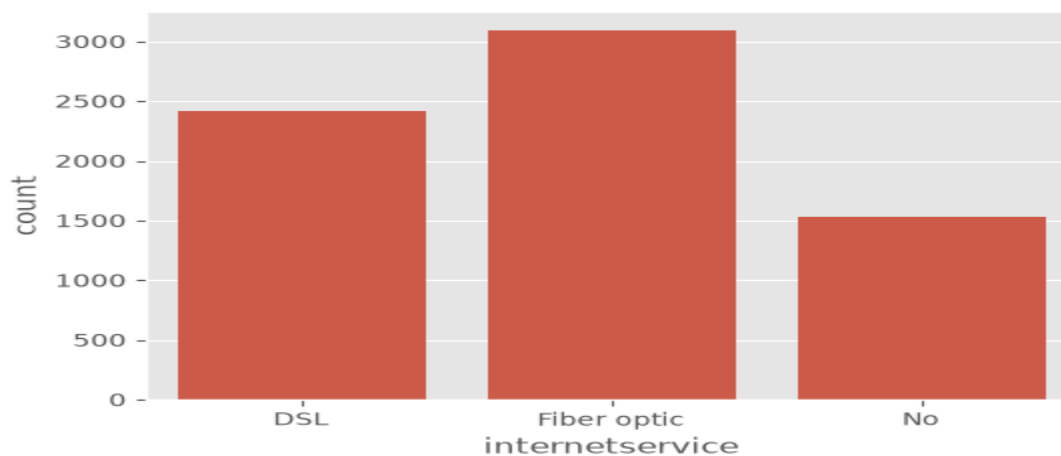
Customer subscriptions to different internet service types were analyzed.

The dataset contains customers subscribed to:

- DSL
- Fiber Optic
- No Internet Service

Observation

- Fiber Optic services have a comparatively higher proportion of churn.
- Customers without internet services generally exhibit lower churn.
- Internet service type significantly influences customer retention behavior.



11. Payment Method Analysis

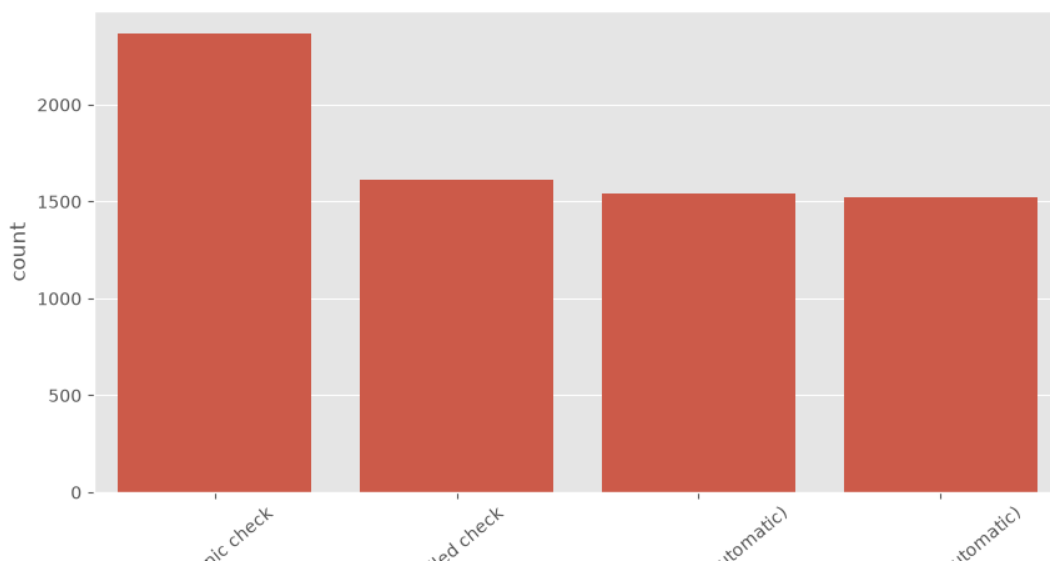
Payment methods were examined to understand customer billing preferences.

The available payment methods include:

- Electronic Check
- Mailed Check
- Bank Transfer
- Credit Card

Observation

- Electronic Check is the most commonly used payment method among churned customers.
- Automatic payment methods appear to be associated with lower churn.
- Payment behavior provides valuable information for customer retention strategies.



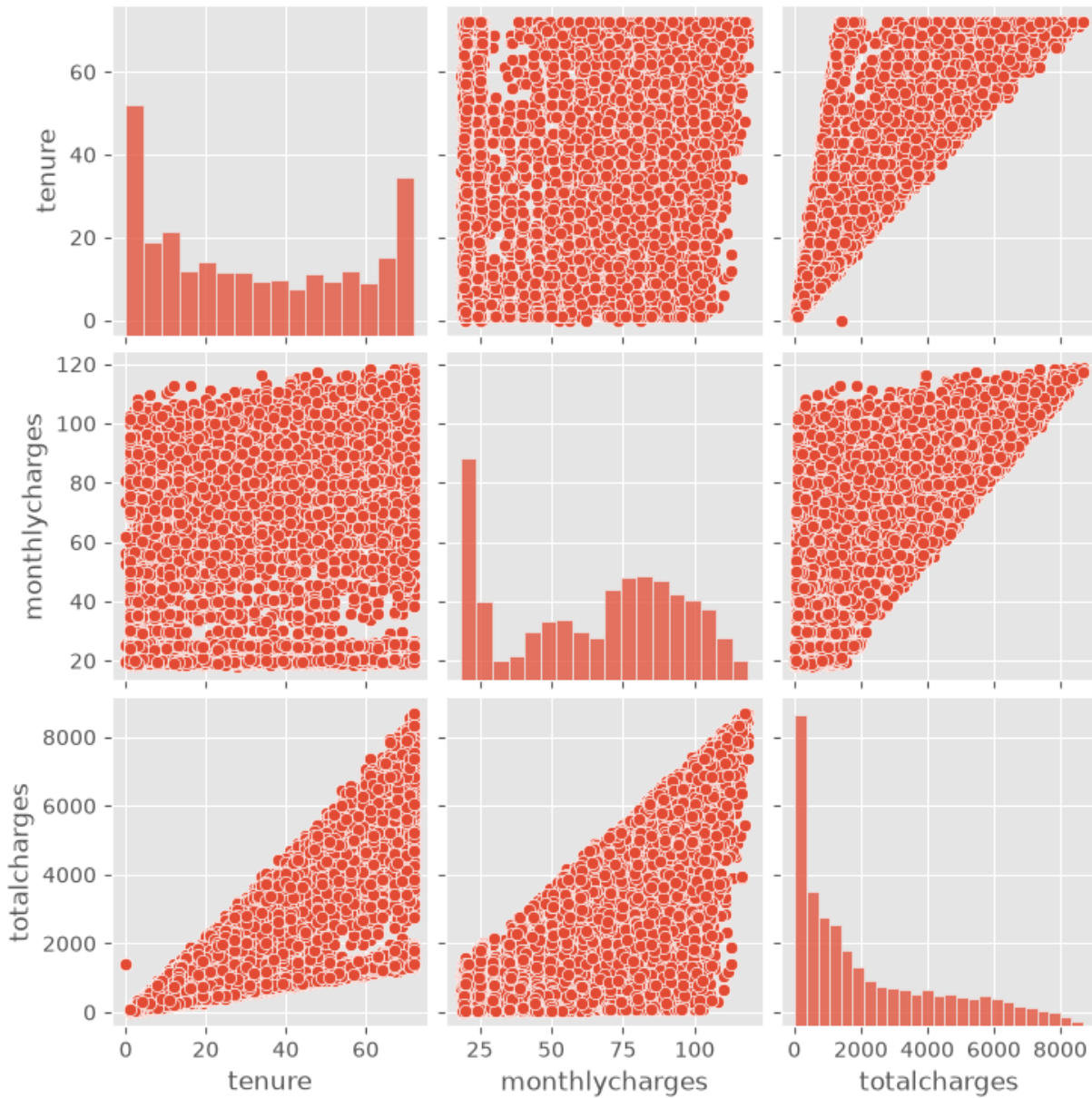
12. Pairplot Analysis

A pairplot was created to visualize relationships among multiple numerical variables simultaneously.

The visualization helps identify clusters, distributions, and pairwise correlations.

Observation

- Strong relationships exist between Total Charges and Tenure.
- Monthly Charges contribute significantly to customer spending patterns.
- The visualization supports later clustering and predictive modeling.



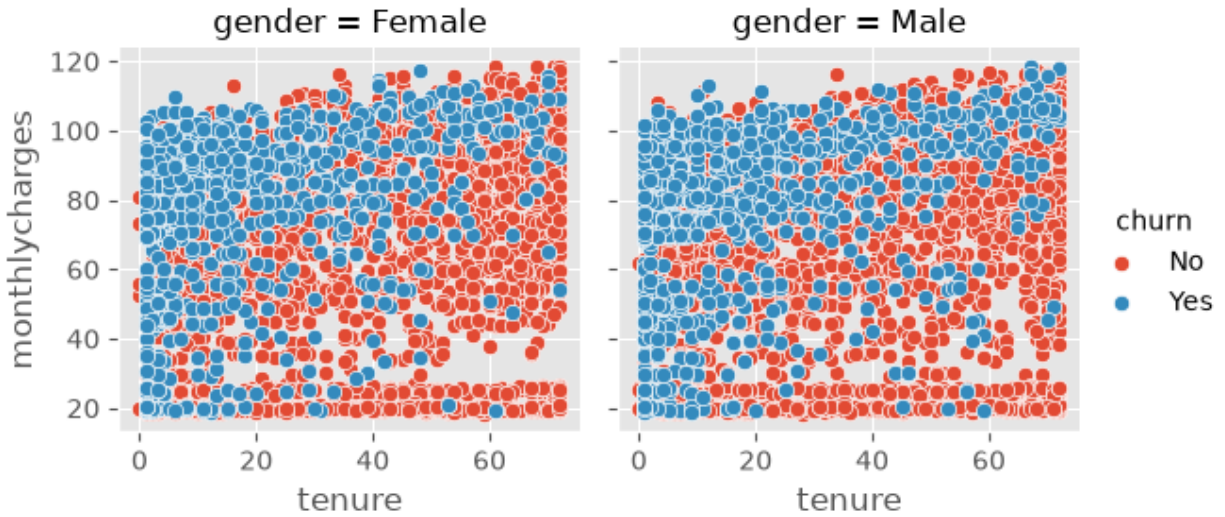
13. FacetGrid Analysis

FacetGrid was used to compare customer behavior across multiple categories simultaneously.

Separate plots were generated based on gender and customer churn.

Observation

- Similar behavioral patterns are observed across genders.
- Customer tenure and monthly charges remain influential regardless of gender.
- Multiple variables interact to influence customer churn.



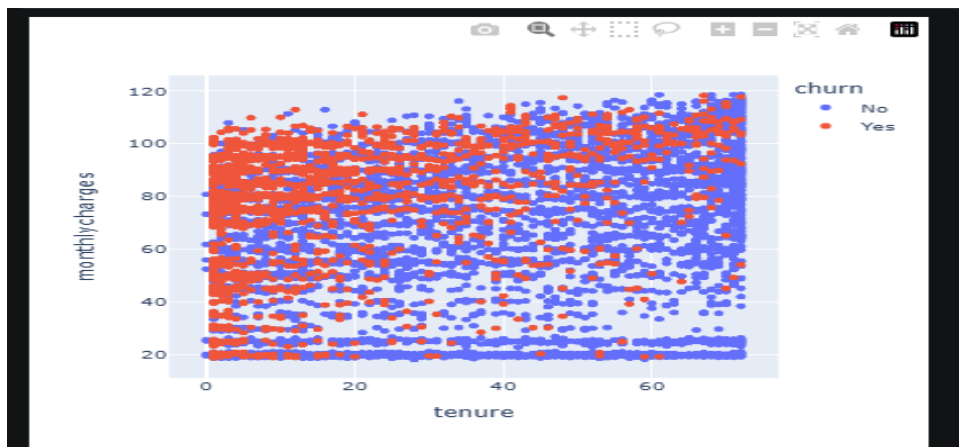
14. Interactive Plotly Visualizations

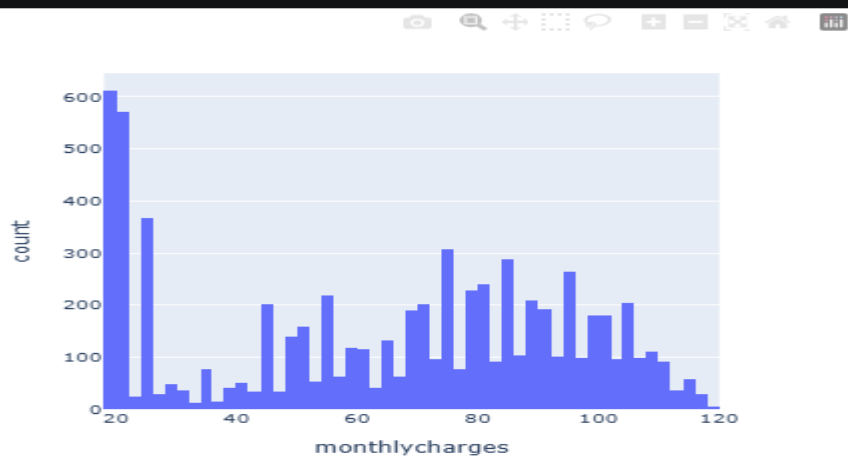
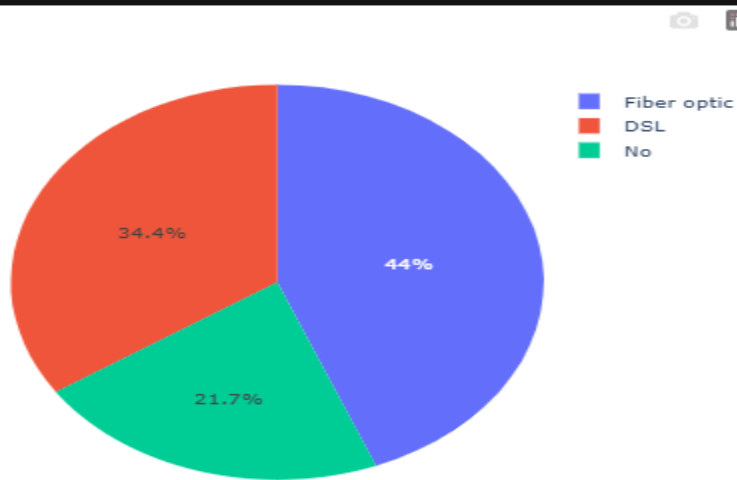
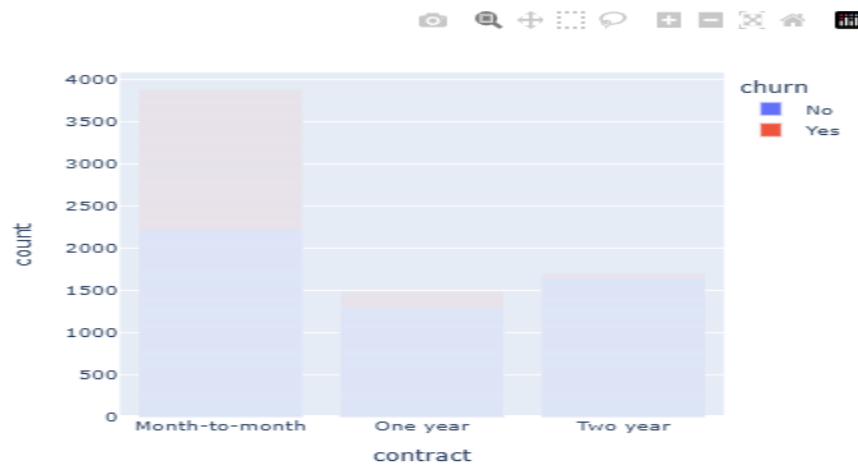
Interactive visualizations were developed using Plotly to enhance data exploration.

The following interactive charts were created:

- Interactive Scatter Plot
- Interactive Bar Chart
- Interactive Pie Chart
- Interactive Histogram

These visualizations allow users to zoom, filter, and interact with customer information, providing a more dynamic analytical experience than static charts.





15. Key Business Insights

The exploratory analysis revealed several important business insights:

- Customers with Month-to-Month contracts experience the highest churn rates.
- Higher Monthly Charges are associated with an increased likelihood of customer churn.
- Customers with longer tenure are significantly less likely to leave the company.
- Fiber Optic customers show comparatively higher churn than customers using DSL or no internet service.
- Electronic Check is the most frequently used payment method among customers who churned.
- Total Charges increase proportionally with customer tenure, indicating stronger customer loyalty over time.
- Gender has minimal influence on customer churn compared to contract type, tenure, and billing characteristics.

16. Conclusion

Exploratory Data Analysis transformed the cleaned customer dataset into meaningful business knowledge. Through statistical summaries and visualizations, hidden patterns influencing customer behavior were identified and interpreted.

The insights generated during this phase established the foundation for the remaining stages of the project, including SQL analysis, interactive dashboard development, statistical forecasting, customer segmentation, and predictive machine learning models.

Overall, EDA demonstrated the importance of visual analytics in understanding customer behavior and supporting data-driven business decision-making.

SQL Analysis & Business Insights

1. Introduction

Structured Query Language (SQL) is one of the most important tools used by data analysts to retrieve, manipulate, and analyze data stored in relational databases. SQL enables organizations to extract meaningful business insights efficiently through filtering, aggregation, grouping, and advanced analytical queries.

As part of this project, the cleaned IBM Telco Customer Churn dataset was imported into a SQLite database. SQL queries were developed to analyze customer behavior, billing patterns, contract information, internet service usage, and churn trends. Python was integrated with SQLite using SQLAlchemy to automate query execution and seamlessly retrieve results into Pandas DataFrames for further analysis.

This phase demonstrates the practical application of SQL in solving real-world business problems and supporting data-driven decision-making.

2. Database Setup

After completing data cleaning and preprocessing, the cleaned dataset (telco_cleaned.csv) was imported into a SQLite database named telco.db.

The database contains a single table named:

customers

Python libraries including SQLite3 and SQLAlchemy were used to establish the database connection and automate data loading.

Database Details

Attribute	Description
Database Name	telco.db
Database Type	SQLite
Table Name	customers
Programming Language	Python
Library Used	SQLite3, SQLAlchemy

The screenshot shows a Jupyter Notebook interface. On the left, a file explorer sidebar displays a directory structure with folders 'data', 'models', and 'notebooks', and a file 'telco.db' under the 'database' folder. The main notebook area shows a code cell with the following Python code:

```
conn = sqlite3.connect("../database/telco.db")
print("database connected successfully")
```

Below the code cell, the output is displayed: `[11] ✓ 0.0s` followed by the text `database connected successfully`.

3. SQL Operations Performed

Several SQL concepts were implemented throughout this project.

Basic SQL Operations

- SELECT
- WHERE
- ORDER BY
- LIMIT

Aggregation Functions

- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()

Grouping Operations

- GROUP BY
- HAVING

Advanced SQL

- Subqueries
- Common Table Expressions (CTEs)
- Window Functions
- ROW_NUMBER()
- RANK()
- LAG()
- LEAD()

Database Optimization

- EXPLAIN QUERY PLAN
- Index Creation

These SQL operations enabled efficient extraction of business information from the customer database.

4. Data Retrieval

The first stage involved retrieving customer records using basic SQL queries.

Typical operations included:

- Displaying all customer records
- Selecting specific columns

- Filtering customers based on churn status
- Sorting customers by monthly charges
- Displaying the first few records using LIMIT

These operations provided an overview of the database contents and verified successful data loading.

```
pd.read_sql("SELECT * FROM customers LIMIT 5", conn)
```

	customerid	gender	seniorcitizen	partner	dependents	tenure	phonservice	multipledlines	internetservice	onlinesecurity	deviceprotection	techsupport	streamingtv	streamingmovies	contract	paperlessbilling	paymentmethod	monthlycharges	totalcharges	churn
0	7890-VHVG	Female	0	Yes	No	1	No	No phone service	DSL	No	—	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1	5575-QNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	—	Yes	No	No	One year	No	Mailed check	56.95	1889.50	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	—	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	—	Yes	Yes	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

5 rows x 21 columns

5. Customer Churn Analysis

Customer churn was analyzed using SQL aggregation queries.

The total number of customers who discontinued the service was calculated using the COUNT() function.

Business Insight

- The company retains the majority of its customers.
- However, the number of churned customers is significant enough to affect long-term revenue.
- Customer retention strategies should focus on identifying high-risk customer segments.

	churn	COUNT(*)
0	No	5174
1	Yes	1869

6. Contract Type Analysis

Customer contracts were analyzed using GROUP BY queries.

The average monthly charges and customer counts were calculated for each contract type.

Business Insight

- Month-to-Month contracts represent the largest customer segment.
- Customers using Month-to-Month contracts exhibit higher churn rates.
- Long-term contracts contribute to stronger customer retention.

These findings indicate that encouraging customers to switch to longer-term contracts may reduce churn.

```

query = """
SELECT contract,
AVG(monthlycharges)
FROM customers
GROUP BY contract;
"""

pd.read_sql(query, conn)
✓ 0.1s

```

	contract	AVG(monthlycharges)
0	Month-to-month	66.398490
1	One year	65.048608
2	Two year	60.770413

7. Payment Method Analysis

SQL queries were executed to analyze customer payment preferences.

Payment methods were grouped and counted using aggregate functions.

Business Insight

- Electronic Check is the most frequently used payment method among churned customers.
- Customers using automatic payment methods appear more likely to remain with the company.

This information can help the company design billing policies that encourage more stable payment methods.

```

query = """
SELECT paymentmethod,
COUNT(*)
FROM customers
GROUP BY paymentmethod;
"""

result = pd.read_sql(query, conn)

result
✓ 0.0s

```

	paymentmethod	COUNT(*)
0	Bank transfer (automatic)	1544
1	Credit card (automatic)	1522
2	Electronic check	2365
3	Mailed check	1612

8. Internet Service Analysis

Customer internet service subscriptions were analyzed using SQL.

The number of customers for each internet service category was calculated.

Business Insight

- Fiber Optic customers represent a large portion of the customer base.
- Fiber Optic users experience comparatively higher churn.
- DSL customers demonstrate relatively better customer retention.

These findings suggest that customer experience for Fiber Optic services should be investigated further.

```
query = """
SELECT internetervice,
AVG(tenure)
FROM customers
GROUP BY internetervice;
"""

result = pd.read_sql(query, conn)

result
```

✓ 0.0s

	internetervice	AVG(tenure)
0	DSL	32.821561
1	Fiber optic	32.917959
2	No	30.547182

9. Aggregate Function Analysis

Aggregate functions were used extensively throughout the project.

The following calculations were performed:

- Total number of customers
- Average monthly charges
- Maximum monthly charges
- Minimum monthly charges
- Total monthly revenue

These metrics provide a high-level summary of customer billing behavior and support executive decision-making.

```
query = """
SELECT COUNT(*)
FROM customers;
"""

result = pd.read_sql(query, conn)

result
✓ 0.0s
```

	COUNT(*)
0	7043

```
query = """
SELECT SUM(monthlycharges)
FROM customers;
"""

result = pd.read_sql(query, conn)

result
✓ 0.0s
```

	SUM(monthlycharges)
0	456116.6

```
query = """
SELECT AVG(monthlycharges)
FROM customers;
"""
```

```
result = pd.read_sql(query, conn)
```

```
result
```

✓ 0.0s

AVG(monthlycharges)	
0	64.761692

```
query = """
SELECT MIN(monthlycharges)
FROM customers;
"""
```

```
result = pd.read_sql(query, conn)
```

```
result
```

✓ 0.0s

MIN(monthlycharges)	
0	18.25

```
query = """
SELECT MAX(monthlycharges)
FROM customers;
"""

result = pd.read_sql(query, conn)

result
```

✓ 0.0s

MAX(monthlycharges)	
0	118.75

10. Subqueries and Common Table Expressions (CTEs)

Advanced SQL techniques were implemented to improve query readability and solve more complex analytical problems.

Subqueries

Subqueries were used to identify customers whose monthly charges exceeded the overall average.

Common Table Expressions (CTEs)

CTEs simplified the analysis of churned customers by creating temporary result sets that could be reused within the same query.

Benefits

- Improved query readability.
- Reduced code duplication.
- Simplified complex business analysis.

```

query = """
WITH churn_customers AS
(
  SELECT *
  FROM customers
  WHERE churn = 'Yes'
)

SELECT *
FROM churn_customers;
"""

result = pd.read_sql(query, conn)

result

```

✓ 0.0s

	customerid	gender	seniorcitizen	partner	dependents	tenure	phonseservice	multiplelines	internetservice	onlinesecurity	deviceprotection	techsupport	streamingtv	streamingmovies	contract	paperlessbilling	paymentmethod	monthlycharges	totallcharges	churn
0	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	—	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
1	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes
2	9306-CZGJC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	—	Yes	No	Yes	Month-to-month	Yes	Electronic check	89.85	820.50	Yes
3	7892-PQOPD	Female	0	Yes	28	Yes	Yes	Yes	Fiber optic	No	—	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	104.80	3048.05	Yes
4	0280-XJGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	—	Yes	No	Yes	Month-to-month	Yes	Bank transfer (automatic)	103.70	5038.30	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1854	1699-HPSEB	Male	0	No	No	12	Yes	No	DSL	No	—	No	Yes	Yes	One year	Yes	Electronic check	59.80	727.80	Yes
1895	8775-CEBBI	Female	0	No	No	9	Yes	No	DSL	No	—	No	No	No	Month-to-month	Yes	Bank transfer (automatic)	64.20	403.35	Yes
1896	6894-LPHLY	Male	1	No	No	1	Yes	Yes	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Electronic check	75.75	75.75	Yes
1987	0638-TSICW	Female	0	No	No	67	Yes	Yes	Fiber optic	Yes	—	No	No	Yes	Month-to-month	Yes	Credit card (automatic)	102.95	6886.25	Yes
1988	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber optic	No	—	No	No	No	Month-to-month	Yes	Mailed check	74.40	306.60	Yes

1869 rows x 21 columns

11. Window Function Analysis

Window functions were implemented to perform advanced analytical calculations.

The following functions were used:

- ROW_NUMBER()
- RANK()
- LAG()
- LEAD()

These functions enable row-level calculations while preserving the original dataset structure.

Business Applications

- Ranking customers by monthly charges.
- Comparing customer billing values.
- Identifying sequential changes in customer data.

Window functions are widely used in modern business analytics and reporting systems.

```

query = """
SELECT customerid,
monthlycharges,

ROW_NUMBER()
OVER(ORDER BY monthlycharges DESC)
AS Rank

FROM customers;
"""

result = pd.read_sql(query, conn)

result

```

✓ 0.0s

	customerid	monthlycharges	Rank
0	7569-NMZYQ	118.75	1
1	8984-HPEMB	118.65	2
2	5989-AXPUC	118.60	3
3	5734-EJKXG	118.60	4
4	8199-ZLLSA	118.35	5
...	...	...	...
7038	0621-CXBKL	18.70	7039
7039	9945-PSVIP	18.70	7040
7040	0827-ITJPH	18.55	7041
7041	9764-REAFF	18.40	7042
7042	6823-SIDFQ	18.25	7043

7043 rows x 3 columns


```

query = """
SELECT customerid,
monthlycharges,

RANK()
OVER(ORDER BY monthlycharges DESC)
AS Rank

FROM customers;
"""

result = pd.read_sql(query, conn)

result

```

[22] ✓ 0.0s

...

	customerid	monthlycharges	Rank
0	7569-NMZYQ	118.75	1
1	8984-HPEMB	118.65	2
2	5989-AXPUC	118.60	3
3	5734-EJXKG	118.60	3
4	8199-ZLLSA	118.35	5
...	...	...	...
7038	0621-CXBKL	18.70	7039
7039	9945-PSVIP	18.70	7039
7040	0827-ITJPH	18.55	7041
7041	9764-REAFF	18.40	7042
7042	6823-SIDFQ	18.25	7043

7043 rows x 3 columns

```

query = """
SELECT customerid,
monthlycharges,

LEAD(monthlycharges)
OVER(ORDER BY tenure)
AS NextCharge

FROM customers;
"""

result = pd.read_sql(query, conn)

result

```

✓ 0.0s

	customerid	monthlycharges	NextCharge
0	4472-LVYGI	52.55	20.25
1	3115-CZMZD	20.25	80.85
2	5709-LVOEQ	80.85	25.75
3	4367-NUYAO	25.75	56.05
4	1371-DWPAZ	56.05	19.85
...	...	...	...
7038	8468-FZTOE	74.35	63.10
7039	2274-XUATA	63.10	104.95
7040	7203-OYKCT	104.95	21.15
7041	2569-WGERO	21.15	103.20
7042	2234-XADUH	103.20	NaN

7043 rows x 3 columns

```

query = """
SELECT customerid,
monthlycharges,

LAG(monthlycharges)
OVER(ORDER BY tenure)
AS PreviousCharge

FROM customers;
"""

result = pd.read_sql(query, conn)

result

```

✓ 0.0s

	customerid	monthlycharges	PreviousCharge
0	4472-LVYGI	52.55	NaN
1	3115-CZMZD	20.25	52.55
2	5709-LVOEQ	80.85	20.25
3	4367-NUYAO	25.75	80.85
4	1371-DWPAZ	56.05	25.75
...	...	...	...
7038	8468-FZTOE	74.35	55.65
7039	2274-XUATA	63.10	74.35
7040	7203-OYKCT	104.95	63.10
7041	2569-WGERO	21.15	104.95
7042	2234-XADUH	103.20	21.15

7043 rows x 3 columns

12. Python and SQL Integration

Python was integrated with SQLite using SQLAlchemy to automate database operations.

The integration included:

- Establishing database connections.
- Executing SQL queries.
- Loading query results into Pandas DataFrames.
- Performing parameterized SQL queries.
- Building reusable database utility functions.

This integration combines the flexibility of Python with the efficiency of SQL.

Advantages

- Automation of repetitive tasks.
- Improved workflow efficiency.
- Simplified data analysis.
- Enhanced scalability.

```
query = """
SELECT customerid,
monthlycharges,

LEAD(monthlycharges)
OVER(ORDER BY tenure)
AS NextCharge

FROM customers;
"""

result = pd.read_sql(query, conn)

result
✓ 0.0s
```

	customerid	monthlycharges	NextCharge
0	4472-LVYGI	52.55	20.25
1	3115-CZMZO	20.25	80.85
2	5709-LVOEQ	80.85	25.75
3	4367-NUYAO	25.75	56.05
4	1371-DWPAZ	56.05	19.85
...	...	...	...
7038	8468-FZTOE	74.35	63.10
7039	2274-XUATA	63.10	104.95
7040	7203-OYKCT	104.95	21.15
7041	2569-WGERO	21.15	103.20
7042	2234-XADUH	103.20	NaN

7043 rows x 3 columns

13. Query Optimization

Database performance was evaluated using the EXPLAIN QUERY PLAN command.

An index was also created on the Contract column to improve query execution speed for frequently used filtering operations.

Optimization techniques become increasingly important when working with large-scale enterprise databases.

```

query = """
EXPLAIN QUERY PLAN

SELECT paymentmethod,
COUNT(*)

FROM customers

GROUP BY paymentmethod;
"""

result = pd.read_sql(query, conn)

result

```

✓ 0.0s

	id	parent	notused	detail
0	6	0	216	SCAN customers
1	8	0	0	USE TEMP B-TREE FOR GROUP BY

14. Key Business Insights

The SQL analysis generated several important business insights:

- Most customers remain with the company, although customer churn remains a significant business challenge.
- Customers with Month-to-Month contracts exhibit the highest churn rates.
- Higher monthly charges are associated with an increased likelihood of customer churn.
- Electronic Check is the most frequently used payment method among customers who discontinued the service.
- Fiber Optic customers demonstrate comparatively higher churn than DSL customers.
- Long-term customers contribute significantly to overall company revenue.
- SQL-based business analysis enables rapid identification of customer segments requiring targeted retention strategies.

15. Conclusion

SQL played a critical role in transforming the cleaned customer dataset into actionable business insights. From basic filtering operations to advanced analytical queries using CTEs and window functions, SQL enabled efficient exploration of customer behavior and billing patterns.

The integration of Python and SQLite further automated data retrieval and analysis, creating a scalable workflow suitable for business intelligence applications.

The findings obtained during this phase directly supported dashboard development, statistical modeling, customer segmentation, and machine learning tasks completed in the subsequent stages of the project.

Dashboard Development (Power BI)

1. Introduction

Business Intelligence (BI) dashboards enable organizations to monitor business performance, identify trends, and support strategic decision-making through interactive visualizations. Unlike static reports, dashboards allow users to filter data, compare metrics, and explore information dynamically.

As part of this project, an interactive Power BI Dashboard was developed using the cleaned IBM Telco Customer Churn dataset. The dashboard combines multiple visualizations into a single interface, enabling users to analyze customer demographics, service subscriptions, billing information, contract types, payment methods, and customer churn efficiently.

The primary objective of the dashboard is to convert analytical results into an intuitive visual format that supports business users and management teams in making data-driven decisions.

2. Dashboard Development Process

The dashboard development process consisted of the following stages:

Step 1 – Data Import

The cleaned dataset (telco_cleaned.csv) was imported into Power BI Desktop using the Get Data → Text/CSV option.

Step 2 – Data Validation

The imported data was verified to ensure:

- Correct data types
- Proper column names
- No missing values
- Consistency with the cleaned dataset

Step 3 – Data Modeling

Relationships between columns were validated, and calculated measures were created where required using Power BI's built-in aggregation features.

Step 4 – Dashboard Design

Multiple visualizations were added and arranged into a user-friendly layout to provide a comprehensive overview of customer behavior and churn patterns.

Step 5 – Interactivity

Interactive slicers and filters were incorporated to allow users to explore the data based on different customer attributes.

3. Dashboard Components

The dashboard consists of several visual elements designed to answer important business questions.

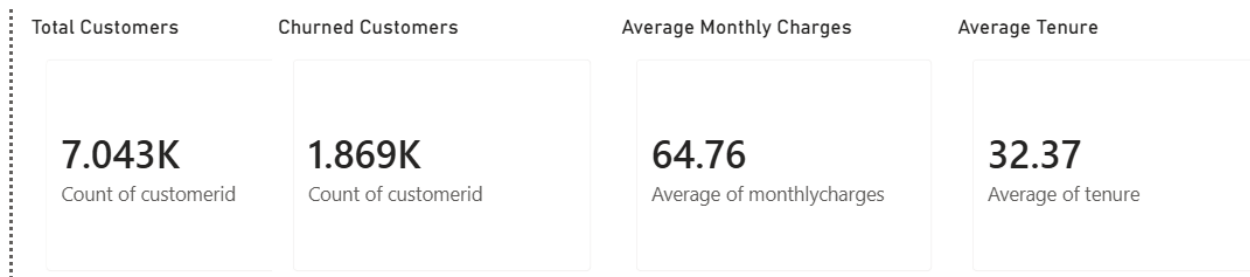
3.1 KPI Cards

Key Performance Indicator (KPI) cards provide a quick summary of important business metrics.

The dashboard includes:

- Total Customers
- Churned Customers
- Average Monthly Charges
- Average Customer Tenure

These KPIs provide management with an immediate overview of customer performance.



3.2 Customer Churn Distribution

A Donut Chart was used to visualize the proportion of customers who remained with the company versus those who churned.

Observation

- The majority of customers continue using the company's services.
- A noticeable percentage of customers have discontinued their subscriptions.
- Churn remains a significant business challenge.

Churn Distribution



3.3 Contract Type Analysis

A Bar Chart was created to analyze customer distribution across different contract types.

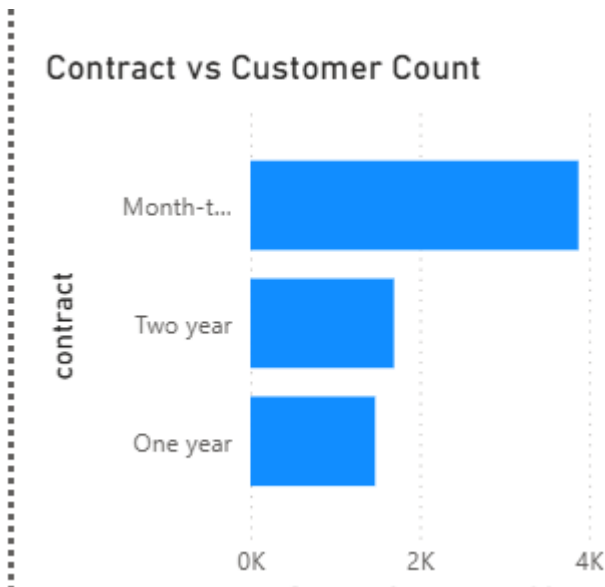
Contract categories include:

- Month-to-Month

- One-Year
- Two-Year

Observation

- Month-to-Month contracts represent the largest customer segment.
- Customers with long-term contracts demonstrate stronger retention.
- Contract type significantly influences customer churn.



3.4 Internet Service Analysis

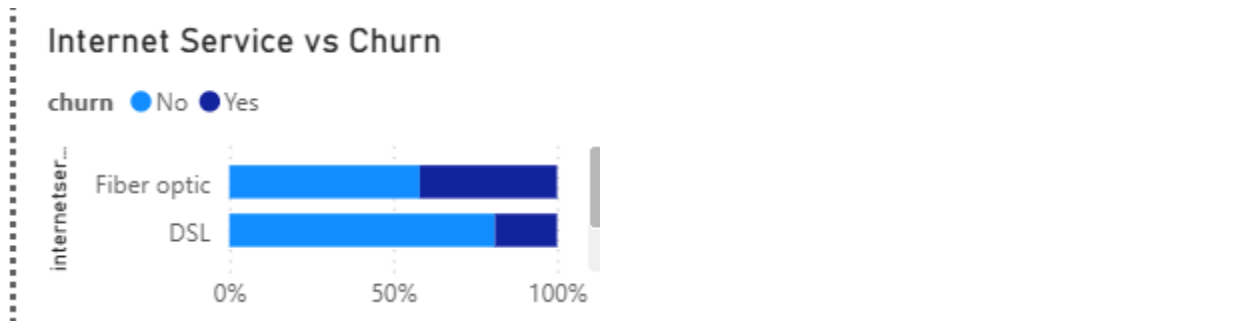
A Stacked Bar Chart was developed to compare customer churn across different internet service types.

The visualization includes:

- DSL
- Fiber Optic
- No Internet Service

Observation

- Fiber Optic customers exhibit comparatively higher churn.
- DSL customers show improved customer retention.
- Customers without internet services have the lowest churn rates.



3.5 Payment Method Analysis

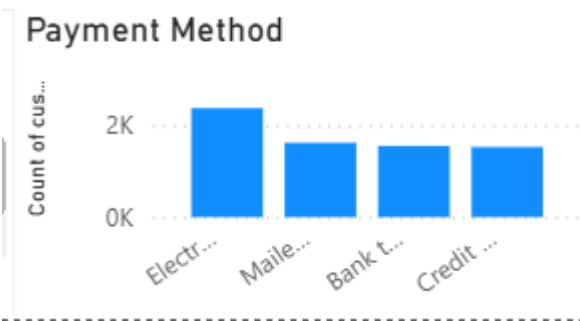
A Clustered Column Chart was used to analyze customer payment preferences.

Payment methods analyzed include:

- Electronic Check
- Mailed Check
- Bank Transfer
- Credit Card

Observation

- Electronic Check is the most common payment method among churned customers.
- Customers using automatic payment methods appear more likely to remain with the company.



3.6 Scatter Plot

A Scatter Plot was created using:

- X-Axis: Customer Tenure
- Y-Axis: Monthly Charges
- Legend: Churn

Observation

- Customers with shorter tenure are more likely to churn.
- Higher monthly charges are associated with increased churn.
- Long-term customers generally demonstrate stronger loyalty.

This visualization supports findings obtained during Exploratory Data Analysis.

Tenure vs Monthly Charges

churn ● No ● Yes



3.7 Matrix and Table Visuals

Matrix and Table visuals provide detailed customer-level information.

Displayed fields include:

- Customer ID
- Contract Type
- Monthly Charges
- Payment Method
- Churn Status

These visuals enable users to perform detailed customer-level analysis directly within the dashboard.

customerid	Sum of monthly	Gender * Churn			
		gender	No	Yes	Total
0002-ORFBO		Female	2549	939	3488
0003-MKNFE		Male	2625	930	3555
0004-TLHLJ		Total	5174	1869	7043
0011-IGKFF					
Total	45€				

3.8 Interactive Slicers

Interactive slicers were added to improve dashboard usability.

Available filters include:

- Gender
- Contract Type
- Internet Service
- Payment Method
- Churn Status

Users can dynamically filter the dashboard and instantly update all visualizations.

This feature significantly enhances decision-making by allowing business users to focus on specific customer segments.

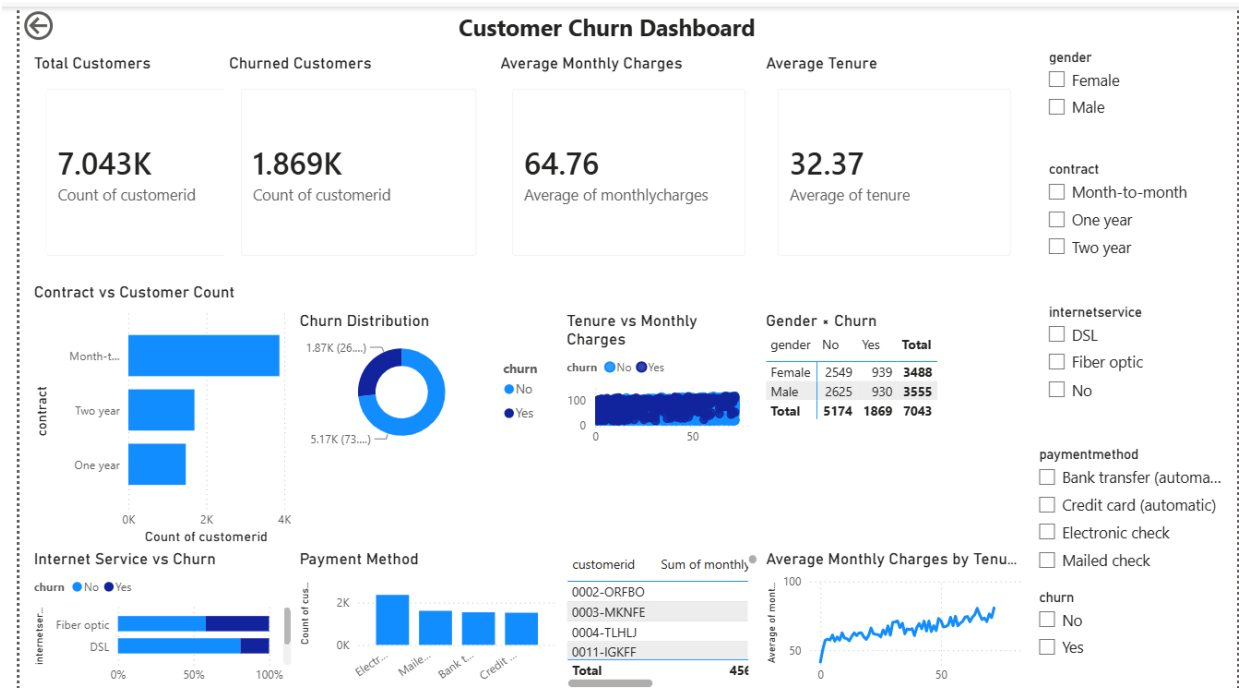
gender☐ Female☐ Male**contract**☐ Month-to-month☐ One year☐ Two year**internetservice**☐ DSL☐ Fiber optic☐ No**paymentmethod**☐ Bank transfer (automa...☐ Credit card (automatic)☐ Electronic check☐ Mailed check

...

...

churn☐ No☐ Yes

...



4. Dashboard Features

The Power BI dashboard includes several advanced features:

- Interactive filtering
- Cross-visual highlighting
- Dynamic KPI updates
- Responsive charts
- Drill-down capability
- Easy navigation
- Business-friendly layout

These features make the dashboard suitable for executive reporting and business intelligence applications.

5. Dashboard Insights

The dashboard revealed several important business insights.

Insight 1

Customers using Month-to-Month contracts have the highest churn rate.

Insight 2

Customers paying higher monthly charges are more likely to discontinue services.

Insight 3

Fiber Optic internet users exhibit comparatively higher churn than DSL customers.

Insight 4

Customers with longer tenure demonstrate significantly lower churn rates.

Insight 5

Electronic Check is the most common payment method among churned customers.

Insight 6

Long-term contracts contribute to improved customer retention.

These insights help management identify customer groups requiring targeted retention strategies.

6. Business Applications

The dashboard can support several real-world business functions, including:

- Customer retention monitoring
- Executive performance reporting
- Marketing campaign analysis
- Customer segmentation
- Billing analysis
- Service quality evaluation
- Strategic decision-making

The interactive nature of the dashboard enables managers to identify trends quickly and respond proactively.

7. Dashboard Performance

The dashboard was designed to ensure:

- Fast data loading
- Efficient filtering
- Clear visualization layout
- Easy interpretation
- Minimal visual clutter

The use of a cleaned dataset improved dashboard responsiveness and overall performance.

8. Dashboard Limitations

Although the dashboard provides valuable business insights, several limitations should be noted:

- The dataset represents historical customer information and does not update in real time.
- Geographic analysis is limited due to the absence of location-specific data.
- Revenue analysis is based on billing information rather than complete financial records.
- Forecasting capabilities are limited by the lack of actual transaction dates.

Future versions of the dashboard could integrate live databases and additional business metrics to enhance functionality.

9. Conclusion

The Power BI dashboard successfully transformed complex customer data into an interactive and visually informative business intelligence solution.

The dashboard integrates key performance indicators, customer demographics, billing analysis, contract information, payment methods, and churn behavior into a single platform, enabling stakeholders to monitor business performance effectively.

By combining interactive filtering, dynamic visualizations, and meaningful business insights, the dashboard serves as a valuable decision-support tool for improving customer retention and enhancing organizational performance.

Overall, the dashboard demonstrates the practical application of Business Intelligence techniques in solving real-world customer analytics problems.

Statistical Analysis & Time Series Forecasting

1. Introduction

Statistical analysis is an essential component of modern data analytics, enabling organizations to understand trends, identify patterns, evaluate data quality, and make informed business decisions. Time series forecasting extends this capability by predicting future observations based on historical trends.

The IBM Telco Customer Churn dataset does not contain actual transaction dates or chronological business records. Therefore, for educational purposes and to demonstrate forecasting techniques, customer tenure (measured in months) was used as a time-like variable. The average monthly charges for customers at each tenure level were calculated to construct a synthetic monthly time series.

This approach allowed the implementation of standard forecasting techniques such as seasonal decomposition, stationarity testing, moving averages, exponential smoothing, and ARIMA modeling while maintaining consistency with the internship requirements.

2. Time Series Construction

The cleaned dataset was grouped by customer tenure, and the average monthly charges were calculated for each tenure value.

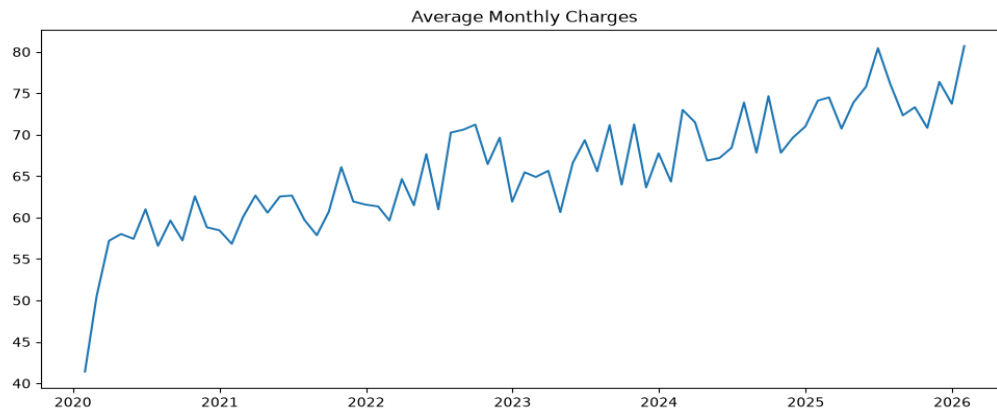
A synthetic monthly timeline was then created using the Date index in Python.

The resulting time series represents the average monthly billing amount across different customer tenure periods.

Steps Performed

- Grouped customer records by tenure.
- Calculated average monthly charges.
- Created a monthly date index.
- Converted the dataset into a time-series format.

This transformed dataset served as the foundation for forecasting analysis.



3. Time Series Visualization

The first step in the analysis involved plotting the time series to observe its overall behavior.

The line chart illustrates how average monthly charges vary across customer tenure.

Observation

- Monthly charges fluctuate across different tenure periods.
- Certain intervals display noticeable increases or decreases.
- Visual inspection suggests the presence of an overall trend with moderate variability.

Time series visualization provides an initial understanding before applying advanced statistical techniques.

4. Seasonal Decomposition

Seasonal decomposition separates a time series into individual components.

The additive decomposition model was applied to divide the series into:

- Trend Component
- Seasonal Component
- Residual Component

Trend

The trend component represents the long-term movement in customer billing behavior.

Seasonal Component

The seasonal component captures repeating patterns within the constructed monthly series.

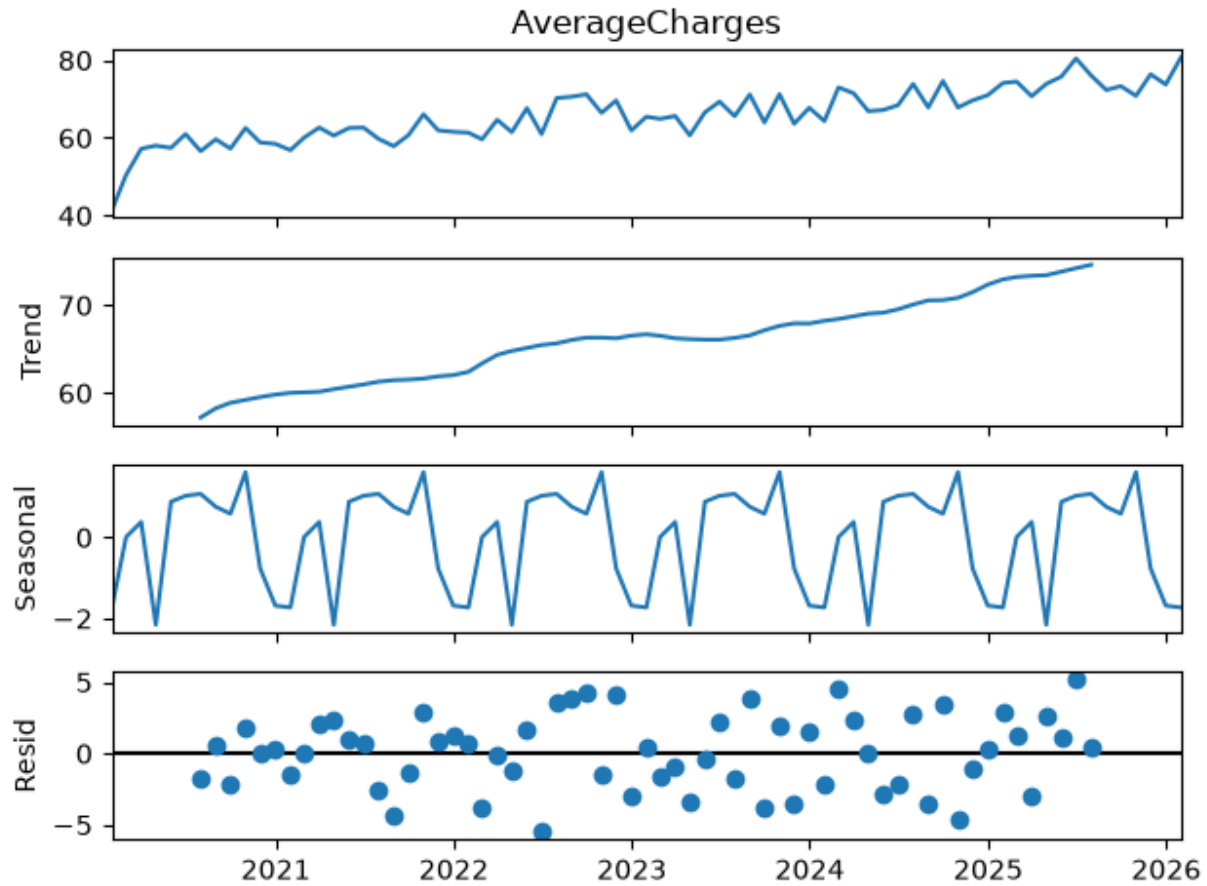
Residual Component

Residual values represent random fluctuations not explained by the trend or seasonality.

Observation

- A smooth underlying trend was observed.
- Seasonal variations were relatively limited due to the nature of the dataset.
- Residual values contained random variations around the trend.

Seasonal decomposition improves understanding of the structure of time-series data before forecasting.



5. Stationarity Testing

Stationarity is an important assumption for many forecasting algorithms, including ARIMA.

The Augmented Dickey-Fuller (ADF) Test was performed to determine whether the constructed time series was stationary.

Hypotheses

Null Hypothesis (H_0): The time series is non-stationary.

Alternative Hypothesis (H_1): The time series is stationary.

The ADF statistic and p-value were calculated using the Statsmodels library.

Interpretation

- If the p-value is less than 0.05, the series is considered stationary.
- Otherwise, differencing or transformation may be required before forecasting.

The ADF test provided statistical evidence regarding the suitability of the data for ARIMA modeling.


```
result = adfuller(ts["AverageCharges"])  
  
print("ADF Statistic:",result[0])  
  
print("p-value:",result[1])
```

```
ADF Statistic: -0.2920082786397437  
p-value: 0.9266285974818688
```

6. Moving Average

Moving averages were calculated to smooth short-term fluctuations and reveal the underlying trend.

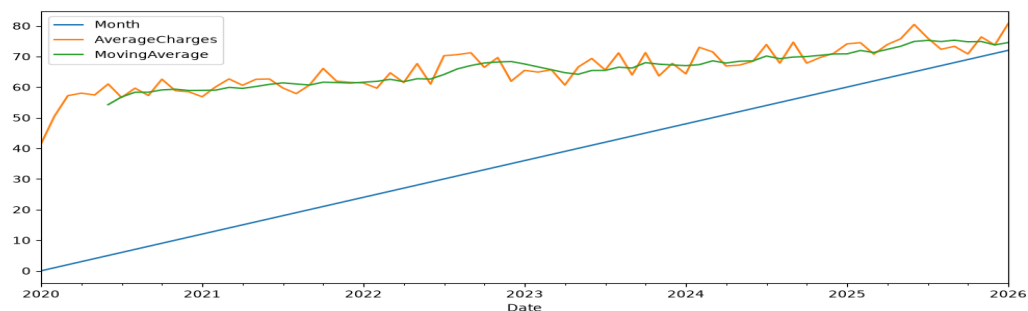
A rolling window was applied to compute the average monthly charges across consecutive observations.

Advantages

- Reduces random fluctuations.
- Highlights long-term trends.
- Simplifies interpretation of customer billing patterns.

Observation

The moving average produced a smoother representation of the original time series, making long-term changes easier to interpret.



7. Exponential Smoothing

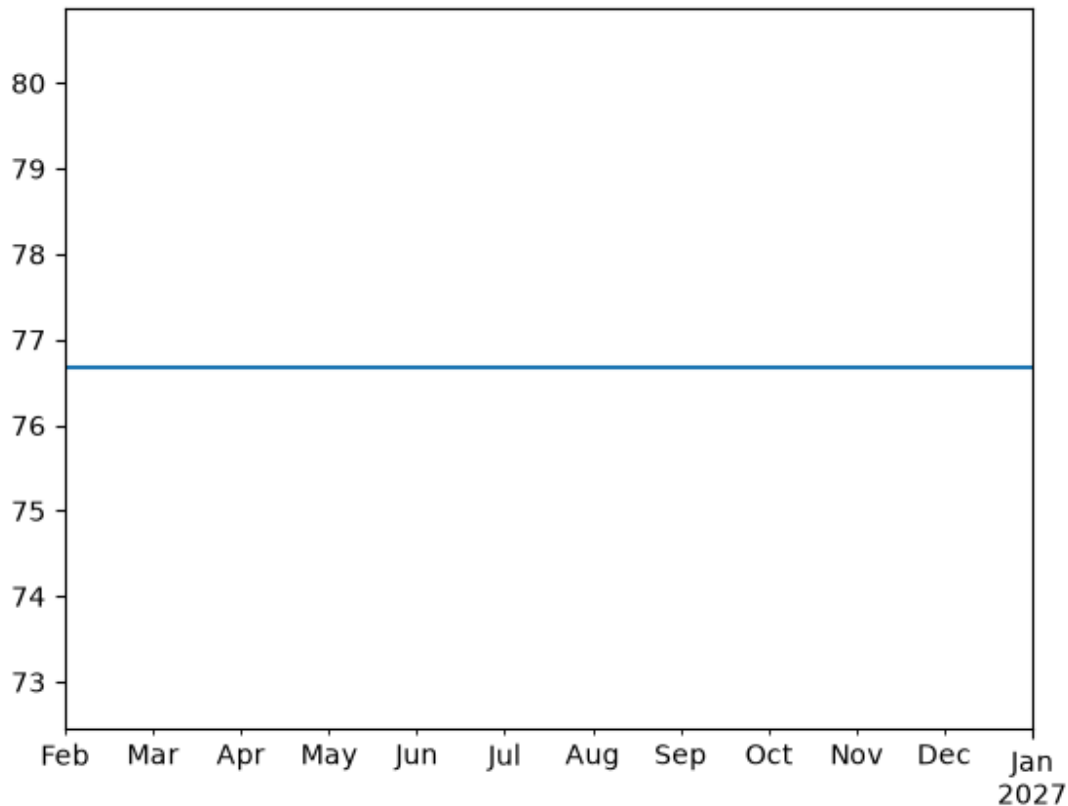
Exponential Smoothing was implemented to generate short-term forecasts.

Unlike the moving average, exponential smoothing assigns greater importance to recent observations while gradually reducing the influence of older data.

Benefits

- Simple implementation.
- Effective for short-term forecasting.
- Reduces random variation.

The resulting forecast follows the recent behavior of the constructed time series while minimizing abrupt fluctuations.



8. ARIMA Forecasting

The AutoRegressive Integrated Moving Average (ARIMA) model was implemented for forecasting.

The ARIMA (1,1,1) configuration was selected to demonstrate forecasting concepts.

The model consists of:

- AutoRegression (AR)

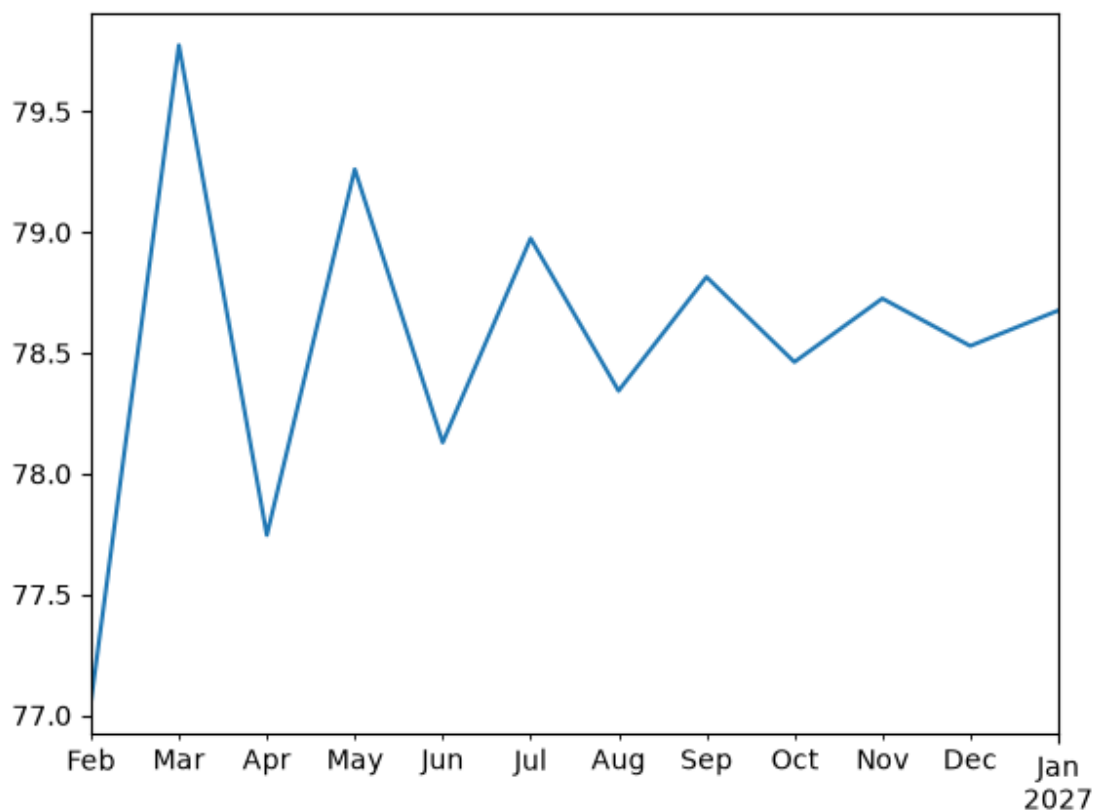
- Integrated Differencing (I)
- Moving Average (MA)

After training, the model generated forecasts for future monthly observations.

Observation

The forecast follows the historical trend while smoothing irregular fluctuations.

Although the dataset is not a true transactional time series, the ARIMA implementation successfully demonstrates the forecasting workflow commonly used in business analytics.



9. Forecast Evaluation

Forecast performance was evaluated using standard regression metrics.

The following metrics were calculated:

Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

A lower MAE indicates better forecasting performance.

Root Mean Squared Error (RMSE)

RMSE measures the magnitude of prediction errors while giving greater weight to larger errors.

A lower RMSE indicates improved forecasting accuracy.

Observation

The calculated evaluation metrics indicate that the forecasting model provides a reasonable approximation of the historical trend within the constructed time series.

```

actual = ts["AverageCharges"][-12:]

predicted = fit.predict()

mae = mean_absolute_error(actual,
                           predicted[-12:])

rmse = sqrt(mean_squared_error(actual,
                                predicted[-12:]))

print(mae)

print(rmse)

```

3.0053350057139787
3.5424323824016177

10. Statistical Analysis Summary

The statistical analysis demonstrated the application of several important forecasting techniques.

The following methods were successfully implemented:

Technique	Status
Time Series Construction	Completed
Time Series Visualization	Completed
Seasonal Decomposition	Completed
ADF Stationarity Test	Completed

Technique	Status
Moving Average	Completed
Exponential Smoothing	Completed
ARIMA Forecasting	Completed
Forecast Evaluation (MAE & RMSE)	Completed

11. Business Applications

The forecasting techniques demonstrated in this project have several practical applications in business environments.

Examples include:

- Revenue forecasting.
- Customer demand prediction.
- Subscription growth estimation.
- Billing trend analysis.
- Capacity planning.
- Resource allocation.
- Financial planning.
- Strategic decision-making.

Organizations can use forecasting models to anticipate future trends and respond proactively to changing business conditions.

12. Limitations

Several limitations should be considered when interpreting the forecasting results.

- The IBM Telco dataset does not contain actual historical transaction dates.
- Customer tenure was used as a surrogate time variable.
- Forecasts demonstrate statistical methodology rather than real operational predictions.
- Additional historical observations would improve forecasting accuracy.

These limitations do not affect the educational value of the analysis but should be acknowledged when interpreting the results.

13. Conclusion

The statistical analysis successfully demonstrated the implementation of modern time-series forecasting techniques using Python and Statsmodels.

Despite the absence of actual chronological business data, the project effectively illustrated the complete forecasting workflow, including time series construction, decomposition, stationarity testing, smoothing techniques, ARIMA modeling, and forecast evaluation.

The knowledge gained from this analysis is directly applicable to real-world forecasting problems involving sales, revenue, customer demand, inventory management, and financial planning.

Overall, this phase strengthens the analytical capabilities demonstrated throughout the project and complements the machine learning and business intelligence components implemented in the later stages of the internship.

Customer Segmentation & Machine Learning

1. Introduction

Machine Learning enables organizations to discover hidden patterns within data, classify customer behavior, predict future outcomes, and support intelligent decision-making. In customer analytics, machine learning techniques are widely used for customer segmentation, churn prediction, revenue estimation, fraud detection, and personalized marketing.

In this project, machine learning techniques were applied to the IBM Telco Customer Churn dataset to identify customer segments and predict customer churn. Three predictive models were developed, namely Linear Regression, Logistic Regression, and Decision Tree Classification. In addition, K-Means Clustering was used to segment customers into meaningful groups based on their service usage and billing characteristics.

The models were developed using the Scikit-learn library and evaluated using industry-standard performance metrics.

2. Customer Segmentation using K-Means Clustering

Customer segmentation is the process of dividing customers into groups with similar characteristics. Segmentation enables organizations to design personalized marketing campaigns, improve customer retention strategies, and optimize service offerings.

In this project, K-Means Clustering was used to group customers based on:

- Customer Tenure
- Monthly Charges
- Total Charges

These variables represent customer loyalty and spending behavior.

3. Feature Scaling

Before applying the clustering algorithm, numerical variables were standardized using the StandardScaler algorithm.

Standardization transforms features so that:

- Mean = 0
- Standard Deviation = 1

Feature scaling is important because K-Means relies on Euclidean distance, and variables with larger numerical ranges can dominate the clustering process if scaling is not performed.

4. Determining the Optimal Number of Clusters

Selecting the appropriate number of clusters is an important step in K-Means clustering.

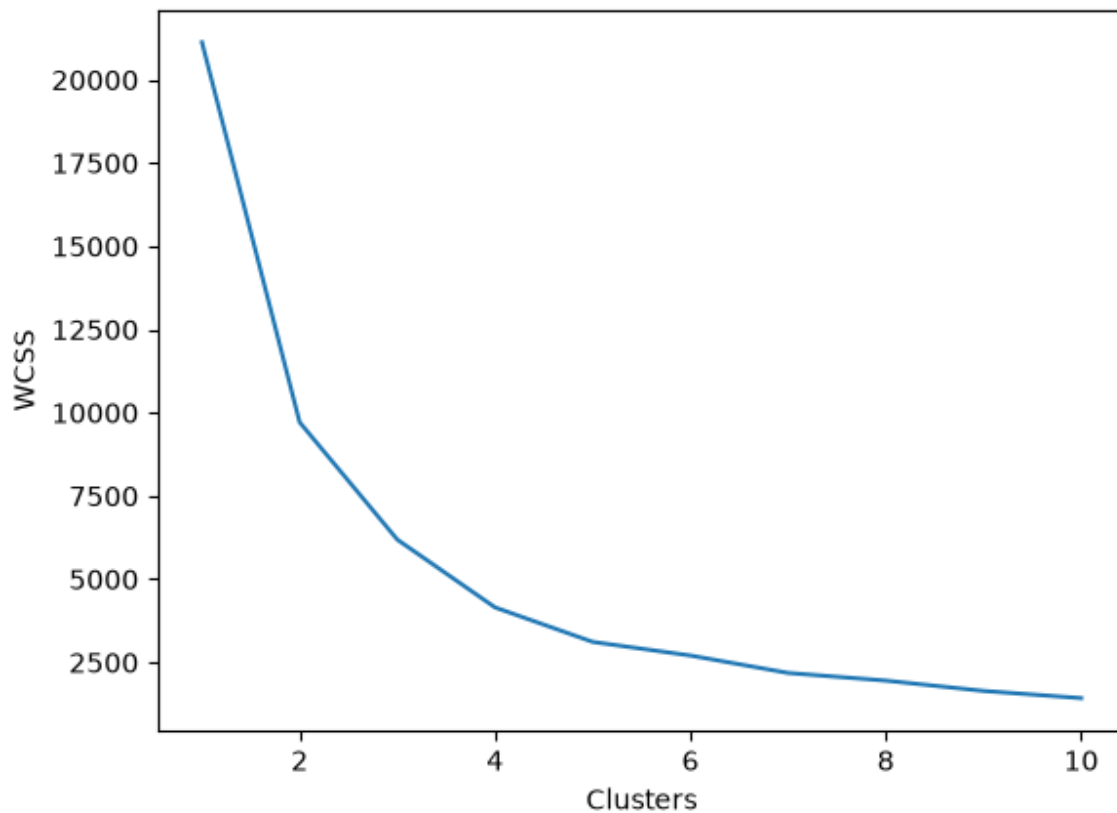
Two evaluation methods were used.

4.1 Elbow Method

The Elbow Method calculates the Within-Cluster Sum of Squares (WCSS) for different values of K.

The optimal value of K is identified where the decrease in WCSS begins to slow significantly.

The analysis suggested that five clusters provided an appropriate balance between simplicity and cluster quality.



4.2 Silhouette Score

The Silhouette Score measures how well each observation fits within its assigned cluster compared to other clusters.

A higher score indicates better-defined and more distinct clusters.

The calculated Silhouette Score confirmed that the selected clustering configuration produced meaningful customer segments.


```
kmeans = KMeans(  
    n_clusters=5,  
    random_state=42  
)  
  
clusters = kmeans.fit_predict(X_scaled)  
  
df["Cluster"] = clusters
```

```
print(  
    silhouette_score(  
        X_scaled,  
        clusters  
    )  
)
```

0.4428557150805072

5. Customer Cluster Interpretation

After applying K-Means, customers were grouped into five clusters.

Each cluster represents customers with similar billing behavior and service usage.

Cluster 1

- Long-tenure customers
- High customer loyalty
- Low churn risk

Cluster 2

- Moderate tenure
- Average monthly charges
- Medium churn probability

Cluster 3

- High monthly charges
- Premium service users
- Higher churn tendency

Cluster 4

- Recently joined customers
- Lower total charges
- Require customer engagement

Cluster 5

- Mixed customer characteristics
- Opportunity for personalized retention strategies

Customer segmentation enables businesses to apply targeted marketing instead of treating all customers equally.

6. PCA Visualization

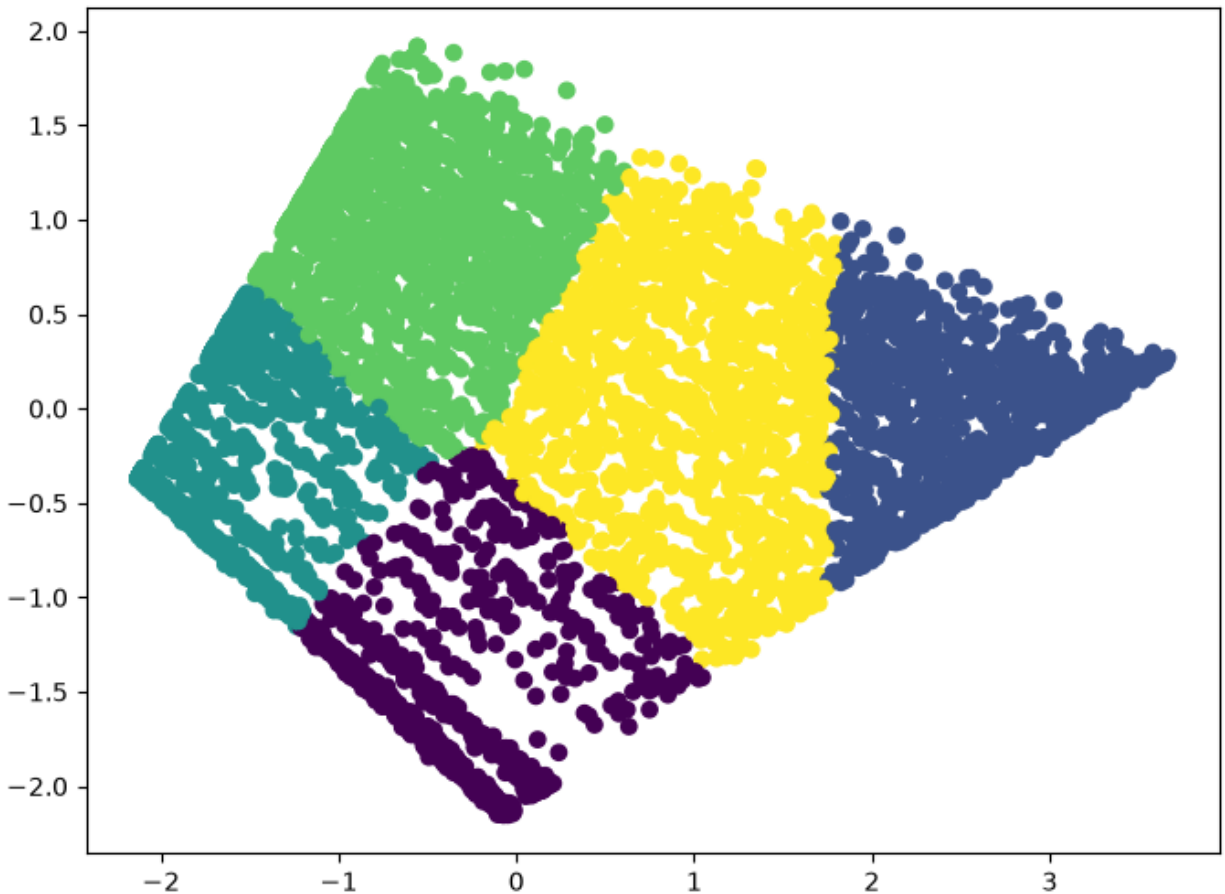
Principal Component Analysis (PCA) was used to reduce the dimensionality of the dataset and visualize customer clusters in two dimensions.

PCA transforms multiple numerical variables into principal components while preserving most of the original information.

The visualization demonstrates clear separation among customer segments.

Observation

- Customer clusters are distinguishable.
- Customers with similar billing patterns appear close together.
- PCA improves interpretability of clustering results.



7. Predictive Modeling

The objective of predictive modeling was to estimate customer behavior using supervised machine learning algorithms.

Three different models were implemented.

8. Linear Regression

Linear Regression was developed to satisfy the internship requirement for numerical prediction.

Instead of predicting customer churn, which is a categorical variable, the model predicts:

Target Variable

Monthly Charges

Input Features

- Customer Demographics
- Contract Information
- Payment Method
- Internet Service
- Tenure

- Total Charges

Model Evaluation

The model was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

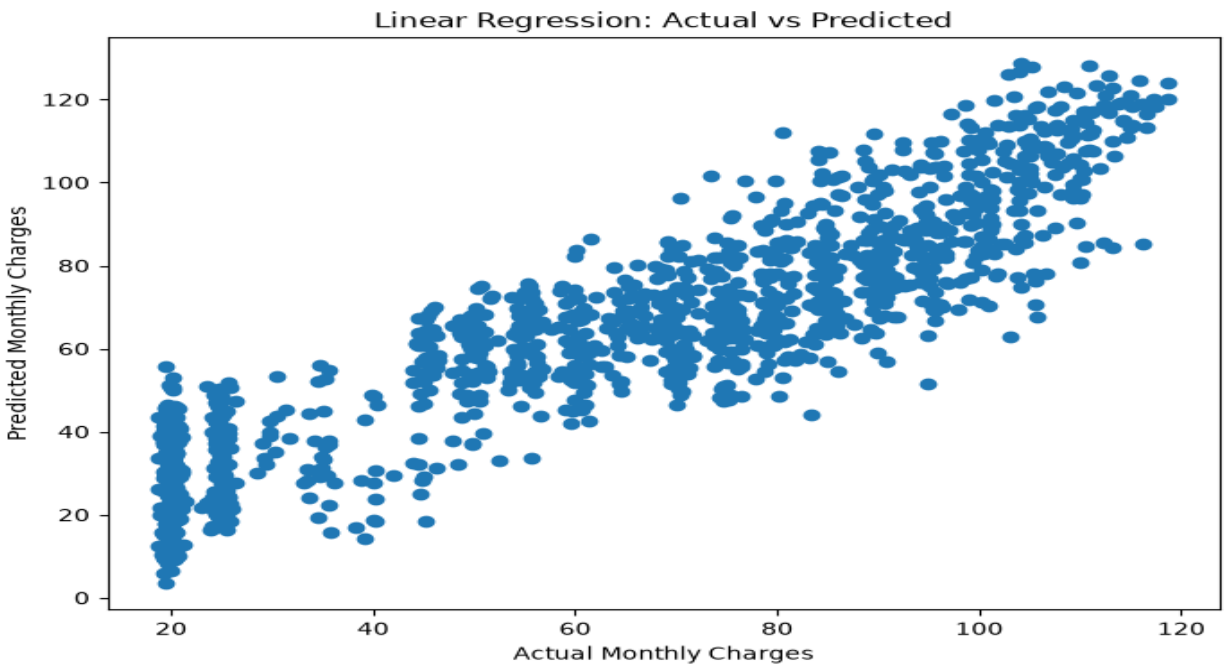
These metrics measure how accurately the model predicts numerical values.

```
Mean Absolute Error : 10.871571411190981
Root Mean Squared Error : 13.231049886629222
R2 Score : 0.8066428231568132
```

Actual vs Predicted Values

A scatter plot comparing actual and predicted Monthly Charges demonstrates the overall predictive capability of the model.

The closer the observations are to the ideal prediction line, the better the model performance.



9. Logistic Regression

Logistic Regression was implemented to predict whether a customer would churn.

Target Variable

Customer Churn

Workflow

- Label Encoding
- Train-Test Split
- Model Training
- Prediction
- Performance Evaluation

The model outputs the probability of customer churn and classifies customers into:

- Churn
- No Churn

Evaluation Metrics

The following classification metrics were calculated:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

These metrics provide a comprehensive evaluation of classification performance.

```
[61] print(classification_report(
      y_test,
      pred
  ))
```

	precision	recall	f1-score	support
0	0.86	0.90	0.88	1036
1	0.68	0.58	0.62	373
accuracy			0.81	1409
macro avg	0.77	0.74	0.75	1409
weighted avg	0.81	0.81	0.81	1409

```
[62] print(
      accuracy_score(
          y_test,
          pred
      )
  )
```

```
0.8147622427253371
```

10. Decision Tree Classification

A Decision Tree classifier was also developed for customer churn prediction.

Decision Trees are widely used because they are highly interpretable and can explain decision-making through a sequence of logical rules.

Advantages

- Easy to interpret
- Handles categorical variables effectively
- Supports feature importance analysis

Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

The Decision Tree results were compared with Logistic Regression to assess model performance.

	precision	recall	f1-score	support
0	0.82	0.81	0.81	1036
1	0.49	0.51	0.50	373
accuracy			0.73	1409
macro avg	0.65	0.66	0.66	1409
weighted avg	0.73	0.73	0.73	1409

11. Confusion Matrix

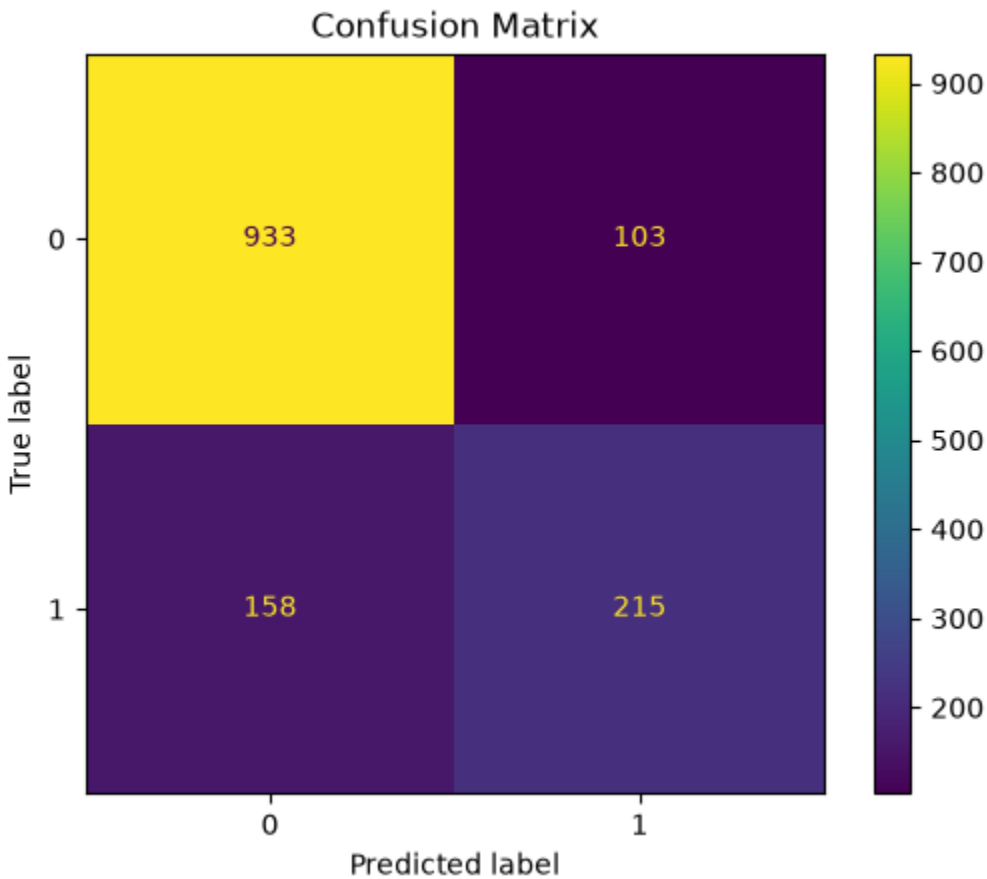
A Confusion Matrix was generated to evaluate the classification performance of the Logistic Regression model.

The confusion matrix summarizes:

- True Positives
- True Negatives
- False Positives
- False Negatives

It provides detailed insight into classification accuracy beyond overall accuracy.

A higher number of correctly classified observations indicates better model performance.



12. ROC Curve

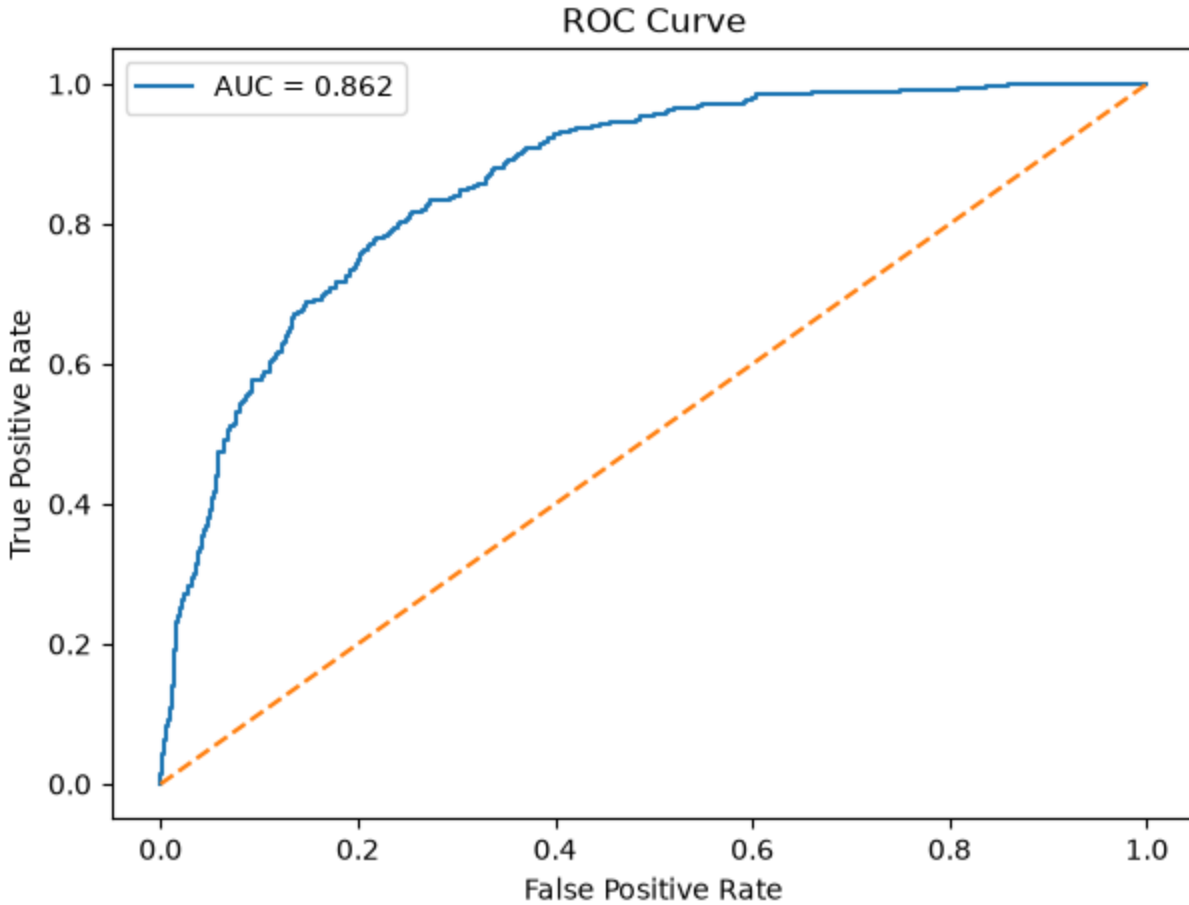
The Receiver Operating Characteristic (ROC) Curve evaluates the model's ability to distinguish between churned and retained customers.

The Area Under the Curve (ROC-AUC) measures classification quality.

Interpretation:

- $AUC = 0.50 \rightarrow$ Random prediction
- $AUC = 1.00 \rightarrow$ Perfect classification

The obtained ROC curve demonstrates that the Logistic Regression model performs significantly better than random guessing.



13. Feature Importance Analysis

Feature Importance analysis identifies the variables that contribute most to model predictions.

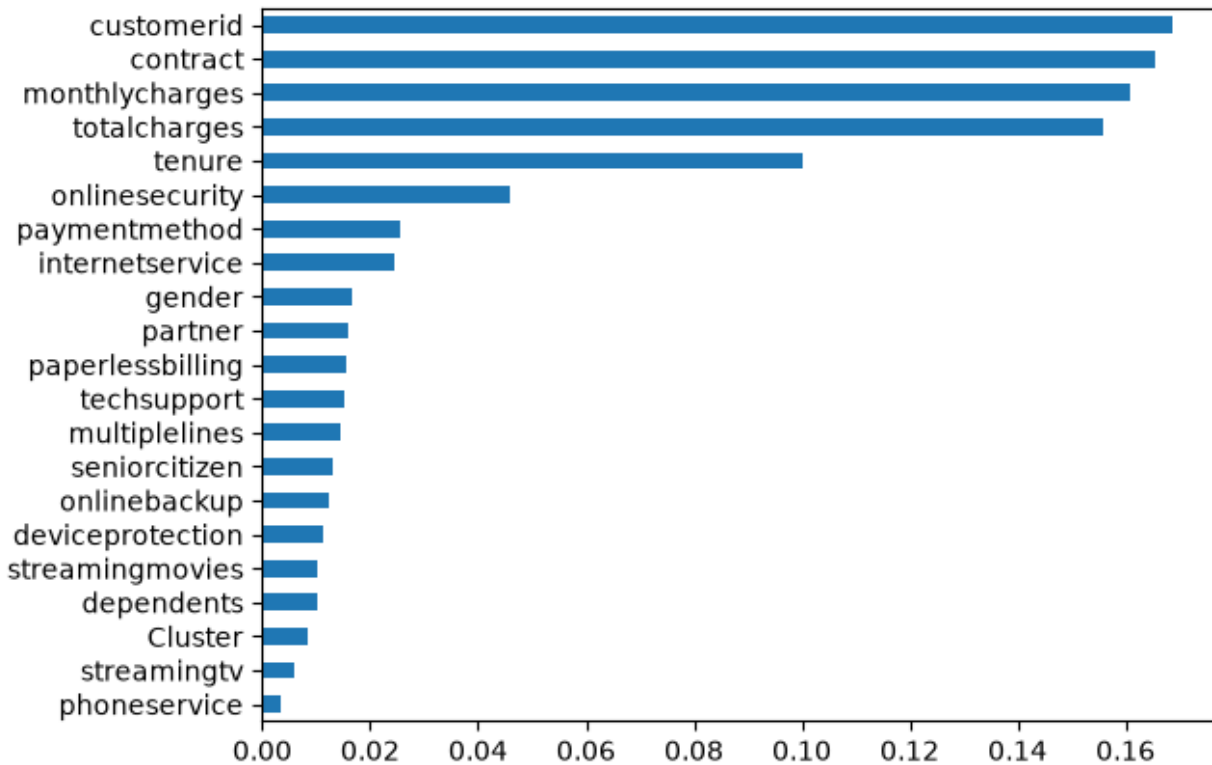
Using the Decision Tree model, feature importance scores were calculated.

The analysis indicates that variables such as:

- Contract Type
- Tenure
- Monthly Charges
- Total Charges

have the greatest influence on customer churn.

These findings are consistent with the observations obtained during Exploratory Data Analysis.



14. Model Comparison

The following machine learning models were implemented during this project.

Model	Purpose	Evaluation Metrics
Linear Regression	Predict Monthly Charges	MAE, RMSE, R ²
Logistic Regression	Predict Customer Churn	Accuracy, Precision, Recall, F1, ROC-AUC
Decision Tree	Predict Customer Churn	Accuracy, Precision, Recall, F1
K-Means Clustering	Customer Segmentation	Elbow Method, Silhouette Score

Each model contributes to different aspects of business decision-making and customer analytics.

15. Business Recommendations

Based on the clustering and predictive modeling results, the following recommendations are proposed:

- Prioritize customer retention efforts for customers identified as high churn risk.
- Encourage Month-to-Month customers to migrate to long-term contracts.
- Design personalized retention campaigns for high-value customers.
- Improve customer support for Fiber Optic service subscribers.
- Promote automatic payment methods to improve customer retention.

- Monitor new customers during their initial months of service.
- Develop loyalty programs for premium customers.

These recommendations can improve customer satisfaction and reduce future churn.

16. Limitations

Although the models performed well, several limitations should be acknowledged.

- The dataset represents historical sample data.
- Hyperparameter tuning was not extensively performed.
- Additional customer behavior variables could improve prediction accuracy.
- External economic and market factors are not included.
- Deep learning models were beyond the scope of this internship.

Future work can incorporate more advanced algorithms such as Random Forest, XGBoost, LightGBM, or Neural Networks.

17. Conclusion

The machine learning phase successfully demonstrated the application of supervised and unsupervised learning techniques for customer analytics.

K-Means clustering effectively segmented customers into meaningful groups, while PCA improved visualization and interpretation of customer segments.

Linear Regression demonstrated numerical prediction techniques, and Logistic Regression and Decision Tree Classification successfully predicted customer churn using customer demographics, contract details, and billing information.

The Confusion Matrix, ROC Curve, Feature Importance analysis, and evaluation metrics confirmed that the developed models provide valuable business insights and can support customer retention strategies.

Overall, this phase illustrates how machine learning can transform historical customer data into actionable intelligence, enabling organizations to make informed, data-driven decisions and improve long-term business performance.

Business Recommendations, Limitations, Future Scope, Conclusion, References & Appendix

1. Business Recommendations

Based on the comprehensive analysis performed throughout this project, several recommendations are proposed to help improve customer retention and support strategic decision-making within the telecommunications business.

Recommendation 1: Encourage Long-Term Contracts

The analysis revealed that customers with Month-to-Month contracts exhibit the highest churn rate compared to customers with One-Year and Two-Year contracts.

Recommended Actions:

- Offer discounts for annual and long-term subscriptions.
- Provide loyalty rewards for customers renewing their contracts.
- Introduce attractive upgrade plans for existing Month-to-Month customers.

Expected Business Benefit:

- Reduced customer churn.
- Increased customer lifetime value.
- Improved revenue stability.

Recommendation 2: Improve Customer Retention Programs

Customer segmentation and predictive modeling identified groups with a higher likelihood of churn.

Recommended Actions:

- Develop personalized retention campaigns.
- Contact high-risk customers before contract renewal.
- Offer targeted promotional packages.
- Conduct customer satisfaction surveys.

Expected Business Benefit:

- Increased customer satisfaction.
- Improved retention rates.
- Lower customer acquisition costs.

Recommendation 3: Review Pricing Strategy

Customers with higher monthly charges demonstrated a greater tendency to discontinue services.

Recommended Actions:

- Introduce flexible pricing plans.
- Bundle premium services at discounted rates.
- Provide customized plans based on customer usage.

Expected Business Benefit:

- Better pricing competitiveness.
- Increased customer satisfaction.
- Reduced churn among premium customers.

Recommendation 4: Enhance Fiber Optic Service Quality

The analysis indicated that customers using Fiber Optic internet services experienced comparatively higher churn.

Recommended Actions:

- Improve network reliability.
- Strengthen customer support services.
- Monitor service quality continuously.
- Reduce service interruptions.

Expected Business Benefit:

- Improved customer experience.
- Increased customer loyalty.
- Higher service retention.

Recommendation 5: Promote Automatic Payment Methods

Customers using Electronic Check were observed to churn more frequently.

Recommended Actions:

- Encourage customers to switch to automatic payment methods.
- Offer cashback or discounts for recurring payments.
- Simplify online payment processes.

Expected Business Benefit:

- Reduced payment-related churn.
- Improved billing efficiency.
- Enhanced customer convenience.

Recommendation 6: Focus on New Customers

Customers with shorter tenure were found to have a higher probability of churn.

Recommended Actions:

- Introduce structured onboarding programs.
- Provide welcome offers.
- Increase customer engagement during the first six months.
- Schedule proactive follow-up communications.

Expected Business Benefit:

- Stronger early customer relationships.
- Reduced first-year churn.
- Improved customer loyalty.

2. Project Limitations

Although the project successfully demonstrates a complete data analytics workflow, several limitations should be acknowledged.

Dataset Limitations

- The IBM Telco Customer Churn dataset is a publicly available sample dataset.
- It does not represent real-time business operations.
- Geographic information is limited.
- Customer satisfaction ratings are unavailable.
- Marketing campaign information is not included.

Time Series Limitations

- The dataset does not contain actual transaction dates.
- Customer tenure was used as a surrogate time variable for forecasting.
- The forecasting results demonstrate methodology rather than operational forecasting.

Machine Learning Limitations

- Models were trained using available dataset features only.
- Extensive hyperparameter tuning was not performed.
- Advanced ensemble techniques such as Random Forest or XGBoost were beyond the scope of this internship.
- External business factors such as competitor pricing and economic conditions were not considered.

Despite these limitations, the project successfully demonstrates practical applications of data analytics and machine learning.

3. Future Scope

This project can be extended in several ways to improve both analytical accuracy and business value.

Real-Time Analytics

- Integrate with live databases.
- Perform continuous dashboard updates.
- Monitor customer behavior in real time.

Advanced Machine Learning

Future work may include:

- Random Forest
- XGBoost

- LightGBM
- CatBoost
- Artificial Neural Networks
- Deep Learning models

These algorithms may improve prediction accuracy and model robustness.

Cloud Deployment

The solution can be deployed using cloud platforms such as:

- Microsoft Azure
- AWS
- Google Cloud Platform

Cloud deployment enables automated data processing and scalable analytics.

Automated Data Pipeline

Future enhancements may include:

- Automated ETL pipelines.
- Scheduled model retraining.
- Automatic dashboard refresh.
- Email reporting.
- Workflow orchestration using Apache Airflow.

Enhanced Dashboard Features

Additional dashboard capabilities could include:

- Real-time KPIs.
- Geographic maps.
- Customer drill-through pages.
- Mobile dashboard support.
- AI-powered forecasting.
- Executive reporting portals.

Customer Experience Analysis

Future datasets may include:

- Customer feedback.
- Complaint history.
- Support ticket information.
- Social media sentiment.
- Customer satisfaction surveys.

These additional variables would improve predictive modeling and customer segmentation.

4. Conclusion

This project successfully demonstrated a complete end-to-end Data Analytics workflow using the IBM Telco Customer Churn dataset as part of the 45-Day Data Analytics Internship at ApexPlanet Software Pvt. Ltd.

The project began with data collection, environment setup, and comprehensive preprocessing to improve data quality. Exploratory Data Analysis (EDA) provided valuable insights into customer demographics, contract types, billing behavior, internet services, payment methods, and customer churn.

SQL was used to perform business-oriented data extraction and analysis through filtering, aggregation, grouping, subqueries, Common Table Expressions (CTEs), and window functions. Python integration further automated database operations and analytical workflows.

Interactive dashboards developed in Power BI transformed analytical findings into business-friendly visualizations, enabling efficient monitoring of key performance indicators and customer behavior.

Advanced statistical analysis demonstrated forecasting concepts using time-series construction, seasonal decomposition, stationarity testing, exponential smoothing, and ARIMA models.

Customer segmentation using K-Means clustering successfully grouped customers into meaningful segments, while Principal Component Analysis (PCA) enhanced visualization and interpretation.

Machine learning models including Linear Regression, Logistic Regression, and Decision Tree Classification successfully demonstrated predictive analytics for customer billing and churn prediction. Model performance was evaluated using industry-standard metrics including Accuracy, Precision, Recall, F1-Score, ROC-AUC, MAE, RMSE, and R^2 .

Overall, this project illustrates the practical application of modern data analytics, business intelligence, statistics, SQL, dashboard development, and machine learning techniques to solve a real-world customer retention problem.

The knowledge and skills acquired during this internship provide a strong foundation for future work in Data Analytics, Business Intelligence, Data Science, and Machine Learning.

5. References

The following resources were used throughout the project:

- IBM Telco Customer Churn Dataset (Kaggle).
- Python Official Documentation – <https://docs.python.org/>
- Pandas Documentation – <https://pandas.pydata.org/>
- NumPy Documentation – <https://numpy.org/>
- Matplotlib Documentation – <https://matplotlib.org/>
- Seaborn Documentation – <https://seaborn.pydata.org/>
- Plotly Documentation – <https://plotly.com/python/>
- Scikit-learn Documentation – <https://scikit-learn.org/>
- Statsmodels Documentation – <https://www.statsmodels.org/>
- SQLAlchemy Documentation – <https://www.sqlalchemy.org/>
- Microsoft Power BI Documentation – <https://learn.microsoft.com/power-bi/>

- SQLite Documentation – <https://www.sqlite.org/>

6. Appendix

The appendix contains supporting materials and project outputs that complement the analysis presented in this report.

Appendix A – Data Cleaning Outputs

- Dataset Preview
- Missing Value Analysis
- Duplicate Check
- Data Type Conversion
- Outlier Detection

Appendix B – Exploratory Data Analysis

- Histograms
- Boxplots
- Heatmap
- Scatter Plot
- Pairplot
- FacetGrid
- Plotly Visualizations

Appendix C – SQL Analysis

- SQL Query Outputs
- Aggregate Functions
- CTE Results
- Window Functions
- Query Optimization

Appendix D – Dashboard

- Full Power BI Dashboard
- KPI Cards
- Charts
- Slicers
- Matrix

Appendix E – Statistical Analysis

- Time Series Plot
- Seasonal Decomposition
- ADF Test

- Moving Average
- ARIMA Forecast

Appendix F – Machine Learning

- Elbow Method
- PCA Visualization
- Linear Regression Results
- Logistic Regression Results
- Decision Tree Results
- Confusion Matrix
- ROC Curve
- Feature Importance

Appendix G – Project Files

- Jupyter Notebooks
- SQL Scripts
- Power BI Dashboard
- Automation Pipeline
- Machine Learning Models
- Requirements File
- README
- GitHub Repository Structure

Acknowledgement

I would like to express my sincere gratitude to ApexPlanet Software Pvt. Ltd. for providing me with the opportunity to participate in the 45-Day Data Analytics Internship. This internship provided valuable practical exposure to data analytics, SQL, business intelligence, statistical analysis, machine learning, and professional project development.

I also extend my appreciation to the mentors, trainers, and everyone who contributed guidance and support throughout this internship. The knowledge and hands-on experience gained during this program have significantly strengthened my technical skills and enhanced my understanding of real-world data analytics workflows.

Finally, I am grateful for the opportunity to work on this project, which has helped me develop confidence in applying analytical techniques to solve business problems and has prepared me for future opportunities in the field of Data Analytics and Data Science.